Skews in the Phenomenon Space Hinder Generalization in Text-to-Image Generation

现象空间中的偏差阻碍了文本到图像生成的泛化能力

> Author

Abstract. The literature on text-to-image generation is plagued by issues of faithfully composing entities with relations. But there lacks a formal understanding of how entity-relation compositions can be effectively learned. Moreover, the underlying phenomenon space that meaningfully reflects the problem structure is not well-defined, leading to an arms race for larger quantities of data in the hope that generalization emerges out of large-scale pretraining. We hypothesize that the underlying phenomenological coverage has not been proportionally scaled up, leading to a skew of the presented phenomenon which harms generalization. We introduce statistical metrics that quantify both the linguistic and visual skew of a dataset for relational learning, and show that generalization failures of text-to-image generation are a direct result of incomplete or unbalanced phenomenological coverage. We first perform experiments in a synthetic domain and demonstrate that systematically controlled metrics are strongly predictive of generalization performance. Then we move to natural images and show that simple distribution perturbations in light of our theories boost generalization without enlarging the absolute data size. This work informs an important direction towards qualityenhancing the data diversity or balance orthogonal to scaling up the absolute size. Our discussions point out important open questions on 1) Evaluation of generated entity-relation compositions, and 2) Better models for reasoning with abstract relations.

摘要 文本到图像生成的文献中充斥着关于忠实地组合具有关系的实体的问题。但缺乏对如何有效学习实体-关系组合的正式理解。此外，有意义地反映问题结构的潜在现象空间没有得到很好的定义，导致了对更大数量数据的军备竞赛，希望泛化从大规模预训练中出现。我们假设潜在的现象学覆盖范围没有按比例扩大，导致呈现的现象出现偏差，从而损害泛化能力。我们引入了统计指标来量化关系学习的数据集的语言和视觉偏差，并表明文本到图像生成的泛化失败是由于不完整或不平衡的现象学覆盖范围直接造成的。我们首先在合成域中进行实验，并证明系统控制的指标对泛化性能具有很强的预测性。然后我们转向自然图像，并表明根据我们的理论，简单的分布扰动可以提高泛化能力，而无需扩大绝对数据量。这项工作为提高数据多样性或平衡质量提供了一个重要的方向，与扩大绝对规模正交。我们的讨论指出了关于 1) 生成实体-关系组合的评估，以及 2) 用于抽象关系推理的更好模型的重要的开放性问题。

Keywords: Text-to-Image · Generalization · Relational Learning

关键词：文本到图像 · 泛化 · 关系学习

1 Introduction

1 绪论

A visual scene is compositional in nature [51]. Atomic concepts, such as entity and texture, are composed via relations [18]. Relations represent abstract functions that are not visually presented, but modulate the visual realization of concepts. For example, consider a scene "*a cat is chasing a mouse*".

视觉场景本质上是组合性的 [51]。诸如实体和纹理之类的原子概念通过关系组合而成 [18]。关系代表抽象函数，这些函数在视觉上没有呈现，但会调节概念的视觉实现。例如，考虑一个场景"一只猫在追一只老鼠"。

It consists of atomic concepts: cat and mouse. "Chasing" defines a relation that is visually realized as certain postures and orientations of the cat and the mouse. Relations take concepts to fill their roles as functions take variables to fill their arguments.

它由原子概念组成：猫和老鼠。“追逐”定义了一种关系，在视觉上表现为猫和老鼠的特定姿势和方向。关系将概念填充到其角色中，就像函数将变量填充到其参数中一样。

This process is known as role-filler binding [2,12,14,41], where fillers are concrete values and roles are abstract positions.

这个过程被称为角色填充绑定 [2,12,14,41]，其中填充物是具体的值，而角色是抽象的位置。

A horse riding an astronaut

A mouse chasing a cat

A pink box is on top of blue box, which is on top of a yellow box, which is on top of a green box.

(a) (b) (c)

Fig. 1: Example images generated by DALL·E3. In all three cases, entities and relations are common but their compositions are uncommon. DALL·E3 tends to (a) compose entities unnaturally, (b) get trapped by the canonical relation, or (c) disregard the requested ordering. These errors are recurring across multiple trials, suggesting that DALL·E3 does not grasp the abstract notion of relations.

图 1：DALL·E3 生成的示例图像。在所有三种情况下，实体和关系都是常见的，但它们的组合是不常见的。DALL·E3 倾向于 (a) 非自然地组合实体，(b) 被规范关系所困，或 (c) 忽略请求的顺序。这些错误在多次试验中反复出现，表明 DALL·E3 并没有掌握关系的抽象概念。

Due to abstractness, relations are always bound to concrete concepts in the space of observations, posting the challenge of truly grasping the abstract function of a relation and using it in generalizable ways, i.e. composing familiar relations with novel concepts. Recently, pre-trained text-to-image models [3, 4, 40] unleash the power of image synthesis with unprecedented fidelity and controllability.

由于抽象性，关系总是与观察空间中的具体概念绑定在一起，这带来了真正理解关系抽象功能并在可泛化方式中使用它的挑战，即用新概念组合熟悉的关系。最近，预训练的文本到图像模型 [3, 4, 40] 以前所未有的保真度和可控性释放了图像合成的力量。

However, as shown by Figure 1, a pre-trained model cannot generate images faithful to the relational constraints upon seeing uncommon entity-relation compositions. This implies that, pre-trained text-to-image models do not represent role-filler bindings independently of the fillers, leading to an important question of **what hinders the learning of generalizable relations**.

然而，如图 1 所示，预训练模型在遇到不常见的实体-关系组合时，无法生成忠实于关系约束的图像。这意味着，预训练的文本到图像模型并没有独立于填充词来表示角色-填充词绑定，这引发了一个重要问题：是什么阻碍了可泛化关系的学习。

This work investigates this question from the data distribution angle. We conjecture that although pre-training ensures massive data quantity, it does not accomplish a proportionally large coverage of unique phenomena. Figure 2 shows our conceptual framework for text-to-image generation, consisting of three distinct components: A text encoder, a visual decoder and a mechanism to communicate between these two spaces.

这项工作从数据分布的角度探讨了这个问题。我们推测，尽管预训练确保了大量的数据，但它并没有实现与之成比例的独特现象覆盖范围。图2展示了我们用于文本到图像生成的的概念框架，它包含三个不同的组件：文本编码器、视觉解码器以及一种在这两个空间之间进行通信的机制。

We formalize the underlying structure of the data as role-filler bindings which nicely capture the compositional connections between data points. We assume that architectural expressivity and pretraining already enable both the text encoder and the visual decoder to distinctly represent fillers and roles in their corresponding spaces.

我们将数据的底层结构形式化为角色-填充绑定，这很好地捕捉了数据点之间的组合连接。我们假设架构表达能力和预训练已经使文本编码器和视觉解码器能够在其各自的空间中分别表示填充和角色。

Based on this assumption, the communication channel becomes the key to task success. We believe the choice of supervision data crucially affects the behavior of the communication channel.

基于此假设，通信通道成为任务成功的关键。我们认为监督数据的选择对通信通道的行为有至关重要的影响。

To this end, we introduce two metrics that quantify the skew of the underlying structure supported by a dataset. These two metrics take into account linguistic notion of roles and visual notion of roles respectively. Our hypothesis is that generalization failure of text-to-image model is a direct result of phenomenological incompleteness or imbalance under our metric.

为此，我们引入了两个指标来量化数据集支持的底层结构的偏斜。这两个指标分别考虑了语言角色和视觉角色的概念。我们的假设是，文本到图像模型的泛化失败是由于我们指标下的现象学不完整或不平衡造成的。

Our experiments in both synthetic images and natural images demonstrate the strong predictive power of our metrics on generalization performance. We also show that findings from pixel diffusion models carry over to latent diffusion models.

我们在合成图像和真实图像上的实验都证明了我们的指标对泛化性能的强大预测能力。我们还表明，像素扩散模型的发现可以推广到潜在扩散模型。

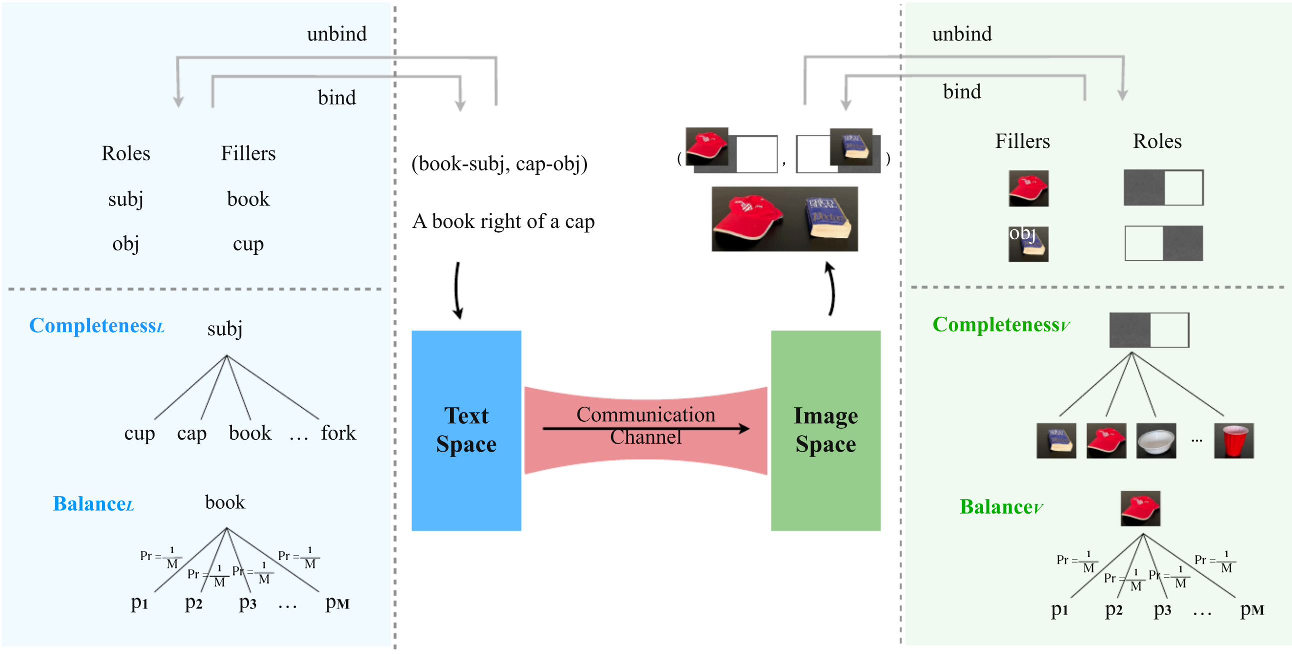


Fig. 2: Conceptual Framework. Text-to-Image generation consists of three important distinct components: A text encoder, a visual decoder, and a mechanism to communicate between these two spaces. Generation of images with consistent spatial relations requires that 1) the text encoder distinctly encodes linguistic roles, 2) the image generator distinguishes spatial roles in the output space, and 3) learning the correct translation from linguistic roles to visual roles. Suppose pre-training or architectural expressivity can fulfill the first two requirements, the remaining core task is to learn an effective communication channel – often instantiated as cross-attention in diffusion models. To this end, we propose statistical metrics to formally quantify how the training data distribution received by the communication channel affects generalization.

图 2：概念框架。文本到图像生成包含三个重要的独立组件：文本编码器、视觉解码器和一种在这两个空间之间进行通信的机制。生成具有一致空间关系的图像需要：1）文本编码器明确地编码语言角色，2）图像生成器区分输出空间中的空间角色，以及 3）学习从语言角色到视觉角色的正确转换。假设预训练或架构表达能力可以满足前两个要求，那么剩余的核心任务是学习一个有效的通信通道——通常在扩散模型中实例化为交叉注意力。为此，我们提出了统计指标来正式量化通信通道接收到的训练数据分布如何影响泛化。

2 Related Work

2 相关工作

Text Conditioned Image synthesis Diffusion models initiate the tide of synthesizing photorealistic images in the wild. They benefit from training stability and do not exhibit mode collapse that GAN models suffer from. [50] feeds text prompts to the diffusion model to make the generation process controllable.

文本条件图像合成扩散模型引领了在野外合成逼真图像的潮流。它们受益于训练稳定性，并且不会出现 GAN 模型所遭受的模式崩溃。[50] 将文本提示馈送到扩散模型，以使生成过程可控。

Inspired by ControlNet, a myriad of works [21, 28, 35, 40, 42, 47] explore the integration of text encoders and image generators, such that image synthesis, editing and style-translation can be customized by users via text. Unlike diffusion models, Transformer-based image synthesis models are naturally better at working in coordination with text, due to the shared tokenization process [4,23].

受 ControlNet 的启发，大量工作 [21, 28, 35, 40, 42, 47] 探索了文本编码器和图像生成器的集成，使得图像合成、编辑和风格转换可以通过文本由用户自定义。与扩散模型不同，基于 Transformer 的图像合成模型由于共享的标记化过程，在与文本协同工作方面自然更胜一筹 [4,23]。

Transformer-based approaches perform on par with diffusion on fidelity, and are believed to have greater potential for resolving long-range dependency and relational reasoning [30], thanks to their patch-based representations and attention blocks. However, Transformers suffer from a discrete latent space and slow inference speed.

基于Transformer的方法在保真度方面与扩散模型相当，并且由于其基于patch的表示和注意力块，被认为在解决长距离依赖和关系推理方面具有更大的潜力[30],。然而，Transformer存在离散潜在空间和推理速度慢的问题。

The latest work [30] integrates a Transformer architecture and diffusion objectives, aiming at the best of both worlds.

最新工作[30]整合了 Transformer 架构和扩散目标，旨在兼顾两者的优势。

Despite high fidelity scores, a recurring problem is the difficulty to generate objects in unfamiliar relations [25]. Although those unfamiliar relations rarely occur in a natural collection of images, they are not physically implausible, and humans have no trouble producing a corresponding scene. This has drawn attention to evaluating generative models and characterizing such failures. Works along this vein suggest that generative models typically fail at multiple objects, with multiple attributes or relations [8, 13], where generalization to novel combinations of familiar components is needed the most.

尽管模型具有高保真度得分，但一个反复出现的问题是难以生成处于不熟悉关系中的物体 [25]。虽然这些不熟悉的关系在自然图像集合中很少出现，但它们在物理上并非不可能，人类也能够轻松地生成相应的场景。这引起了人们对评估生成模型和描述此类失败的关注。沿着这条思路的研究表明，生成模型通常在包含多个物体、多个属性或关系 [8, 13] 的情况下失败，而这些情况下最需要对熟悉组件的新组合进行泛化。

Compositional generalization in image synthesis Compositional generalization is a specific form of generalization where individually known components are utilized to generate novel combinations. This remarkable learning ability has been widely studied in the vision-and-language understanding domain [15, 19, 22, 32, 38, 39, 44].

图像合成中的组合泛化 组合泛化是一种特殊的泛化形式，其中利用已知的单个组件来生成新的组合。这种非凡的学习能力在视觉和语言理解领域得到了广泛的研究 [15, 19, 22, 32, 38, 39, 44。

The text-to-image literature has recently seen efforts towards constructing images compositionally [7, 45, 49]. Closest to ours are two previous works that characterize properties of the underlying phenomenon space not trivially revealed by the pixel space. [29] has investigated shape, color and size as domains of atomic components, which can be combined to form tuples, e.g. (big, red, triangle). They mainly argue that generalization occurs under two conditions: 1) small structural distance between training and testing instances, and 2) effectively learning the disentanglement of attributes (i.e. a change in the size input will not affect the color output).

文本到图像文献最近出现了构建图像的组合方法[7, 45, 49]。与我们的工作最接近的是两项先前的工作，它们描述了底层现象空间的属性，这些属性并非像素空间所能轻易揭示。[29]研究了形状、颜色和大小作为原子组件的域，这些组件可以组合形成元组，例如（大，红色，三角形）。他们主要认为泛化发生在以下两种情况下：1）训练实例和测试实例之间的结构距离较小，以及 2）有效地学习属性的解耦（即，大小输入的变化不会影响颜色输出）。

[43] first assumes compositional data are formed by combining individually complex components with simple aggregation functions. Then they defined *compositional support* and *sufficient support* over a set of components, which are sufficient conditions for a learning system that compositionally generalizes.

[43] 首先假设组合数据是通过将单个复杂组件与简单的聚合函数组合而成的。然后，他们定义了组件集上的组合支持和充分支持，这是组合泛化学习系统的充分条件。

Motivated by failures of existing methods, we investigate data-related factors that affect the generalization performance. There is a possibility that better architectural design can complement high-quality data to achieve generalization.

受现有方法失败的启发，我们研究了影响泛化性能的数据相关因素。有可能更好的架构设计可以补充高质量数据，以实现泛化。

3 Formalization

3 正式化

We start by formalizing scene construction as role-filler bindings. A scene is constructed by binding fillers denoted as $\mathbf{F} = (f_1, \ldots, f_K)$ to roles denoted as $\mathbf{R} = (r_1, \ldots, r_K)$. Roles and fillers are paired up by their indices.

我们首先将场景构建形式化为角色填充绑定。场景通过将表示为 $\mathbf{F} = (f_1, \ldots, f_K)$ 的填充项绑定到表示为 $\mathbf{R} = (r_1, \ldots, r_K)$ 的角色来构建。角色和填充项通过其索引配对。

Hence, each scene representation involves the same number of roles and fillers, i.e., $|\mathbf{F}| = |\mathbf{R}| = K$. Using ψ to denote the *binding* operation, a scene can be formalized into: $\psi(\mathbf{F}, \mathbf{R}) = (f_1/r_1, ..., f_K/r_K)$ and we call (f_k, r_k) a role-filler pair.

因此，每个场景表示都包含相同数量的角色和填充物，即 |F| = |R| = K。使用 ψ 表示绑定操作，一个场景可以形式化为：$\psi(\mathbf{F}, \mathbf{R}) = (f_1/r_1, ..., f_K/r_K)$，我们称 (f_k, r_k) 为角色-填充对。

We would *unbind* a filler from $\psi(\mathbf{F}, \mathbf{R})$ via the unbinding operation ψ^{-1}: $\psi^{-1}\Big(\psi(\mathbf{F}, \mathbf{R}), f_k\Big) = r_k$. The unbinding operation describes the process of extracting the role from a binding that a given filler has been bound to, which corresponds to the decomposition of a compositional structure.

我们将通过解绑操作 ψ^{-1}：$\psi^{-1}\Big(\psi(\mathbf{F}, \mathbf{R}), f_k\Big) = r_k$ 从 ψ(F, R) 中解绑一个填补项。解绑操作描述了从一个给定填补项所绑定到的绑定中提取角色的过程，这对应于组合结构的分解。

Assume that ψ^{-1} returns *null* if the input filler f_k does not exist in the scene.

假设如果输入填充物 f_k 不存在于场景中，则 ψ^{-1} 返回 null。

Fillers are atomic concepts that can be selected from a set of concepts $\mathcal{C} = \{c_1, ..., c_\mathsf{N}\}$ while roles can take values from a set of candidate positions $\mathcal{P} = \{p_1, \ldots, p_\mathsf{M}\}$.

填充物是从概念集 $\mathcal{C} = \{c_1, ..., c_\mathsf{N}\}$ 中选择的原子概念，而角色可以从候选位置集 $\mathcal{P} = \{p_1, \ldots, p_\mathsf{M}\}$ 中取值。

Typically, roles can have intrinsic meanings independent of the meanings of fillers [41]. For instance, the meaning of “upper position" is determined by the y-axis in a 2D image coordinate system, which exists independently of the specific pixel values that fulfill this role in each image.

通常，角色可以具有独立于填充物含义的内在含义 [41]。例如，“上部位置”的含义由二维图像坐标系中的 y 轴决定，它独立于每个图像中满足此角色的特定像素值而存在。

The meanings of roles can be either learned from the task structure or manually designed. In the text-to-image case, the task structure naturally invites two ways to define roles, corresponding to the linguistic and the visual space, respectively. From the linguistic perspective, we consider grammatical positions, e.g. $\mathcal{P}_L = \{subject, object\}$.

角色的含义可以从任务结构中学习，也可以手动设计。在文本到图像的情况下，任务结构自然地引出了两种定义角色的方式，分别对应于语言空间和视觉空间。从语言的角度来看，我们考虑语法位置，例如 $\mathcal{P}_L = \{subject, object\}$。

From the visual perspective, we consider spatial positions, e.g. $\mathcal{P}_V = \{top, bottom\}$.

从视觉角度来看，我们考虑空间位置，例如 $\mathcal{P}_V = \{top, bottom\}$。

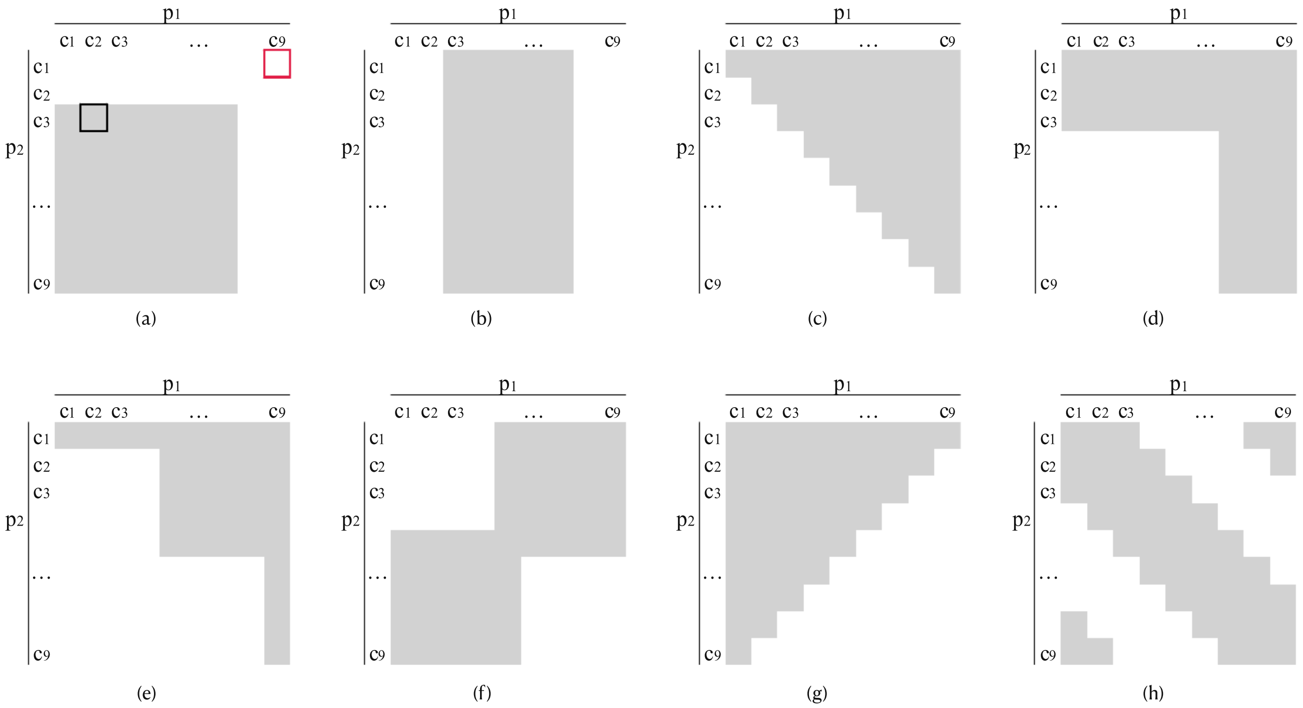


Fig. 3: Sketched illustrations of phenomenological coverage with different properties. Shaded areas represent the training set, while blank areas represent the testing set. Columns and rows are organized by the concepts bound to position 1 and position 2 respectively. For example, the black cell in (a) represents the training instance $(c_2/p_1, c_3/p_2)$, the red cell in (a) represents the testing instance $(c_9/p_1, c_1/p_2)$. (a) Both positions are incomplete (b) Only p_1 is incomplete (c) Complete but unbalanced (d) Complete but unbalanced (e) Complete but unbalanced (f) Complete and balanced (g) Complete and balanced (h) Complete and balanced

图 3：具有不同属性的现象覆盖的示意图。阴影区域代表训练集，空白区域代表测试集。列和行分别按绑定到位置 1 和位置 2 的概念进行组织。例如，(a) 中的黑色单元格代表训练实例 $(c_2/p_1, c_3/p_2)$，(a) 中的红色单元格代表测试实例 $(c_9/p_1, c_1/p_2)$ 。(a) 两个位置都不完整 (b) 只有 p_1 不完整 (c) 完整但失衡 (d) 完整但失衡 (e) 完整但失衡 (f) 完整且平衡 (g) 完整且平衡 (h) 完整且平衡

Under our definitions, each image is a scene. Therefore, an image dataset can be essentially abstracted as a collection of bindings: $\mathcal{D} = \{\psi(\mathbf{F}^i, \mathbf{R}^i)\}_{i=1,\ldots,|\mathcal{D}|}$, where $\mathbf{F}^i$ and $\mathbf{R}^i$ denotes the fillers and roles in the i-th image. Let $\mathcal{U} = \mathcal{C} \times \mathcal{P}$ be the universe of all possible bindings.

根据我们的定义，每张图像都是一个场景。因此，一个图像数据集本质上可以抽象为一个绑定集合：$\mathcal{D} = \{\psi(\mathbf{F}^i, \mathbf{R}^i)\}_{i=1,\ldots,|\mathcal{D}|}$ ，其中 $\mathbf{F}^i$ 和 $\mathbf{R}^i$ 表示第 i 张图像中的填充物和角色。令 U = C × P 为所有可能绑定的全集。

The vanilla notion of coverage supported by a dataset is the proportion of $\mathcal{U}$ that has non-zero supporting examples in the dataset: $\textbf{Coverage}(\mathcal{D}) = |Deduplicate(\mathcal{D})|/|\mathcal{U}|$, where the $Deduplicate$ removes examples with equivalent role-filler representations. We argue that this metric overlooks how elements in $\mathcal{U}$ are structurally connected.

数据集支持的香草覆盖率概念是指在数据集中具有非零支持示例的 $\mathcal{U}$ 的比例: Coverage(D) = |Deduplicate(D)|/|U|，其中 Deduplicate 去除具有等效角色填充表示的示例。我们认为，该指标忽略了 $\mathcal{U}$ 中元素的结构连接方式。

For instance, each element in $\mathcal{U}$ shares a common role or filler with other elements. Without taking this structural property into account, truly meaningful coverage of diverse and unique phenomena might be conflated by the seemingly diverse surface forms.

例如，$\mathcal{U}$ 中的每个元素都与其他元素共享一个共同的角色或填充物。如果不考虑这种结构属性，真正有意义的多样化和独特现象的覆盖范围可能会被看似多样化的表面形式混淆。

Motivated by this consideration, our proposed metrics aim to measure whether a dataset has support for every concept occurring in every position, as well as the probability distribution of the positions that each concept has been bound to. Next, we formally describe completeness and balance metrics by conveniently leveraging the notations of binding and unbinding operators.

受此考虑的启发，我们提出的指标旨在衡量数据集是否支持每个位置出现的每个概念，以及每个概念绑定到的位置的概率分布。接下来，我们通过方便地利用绑定和解绑运算符的符号，正式描述完整性和平衡性指标。

3.1 Completeness

3.1 完备性

Completeness requires that every relation has been bound with every concept across the entire dataset. In this sense, we define:

完整性要求每个关系在整个数据集上与每个概念绑定。从这个意义上说，我们定义：

$$\textbf{Completeness}(p_m, \mathcal{D}) = \frac{\left|\left\{c_n \,\middle|\, p_m \in \psi^{-1}(\mathcal{D}, c_n), c_n \in \mathcal{C}\right\}\right|}{|\mathcal{C}|}, \tag{1}$$

where the operator extracting a concept from a dataset is defined as:

其中从数据集提取概念的操作符定义为：

$$\psi^{-1}(\mathcal{D}, c_n) = \bigcup_{i=1}^{|\mathcal{D}|} \left\{\psi^{-1}\left(\psi(\mathbf{F}^i, \mathbf{R}^i), c_n\right)\right\}. \tag{2}$$

A fully complete dataset should have completeness scores of 1 for all relations. We take the expected value to obtain an aggregated score over the entire dataset:

一个完整的完整数据集应该在所有关系中都具有 1 的完整性得分。我们取期望值来获得整个数据集的聚合得分：

$$\begin{aligned}\textbf{Completeness}(\mathcal{D}) &= \mathbb{E}\left[\textbf{Completeness}(p_m, \mathcal{D})\right] \\ &= \sum_{p_m \in \mathcal{P}} \mathbb{P}(p_m)\textbf{Completenss}(p_m, \mathcal{D}).\end{aligned} \tag{3}$$

3.2 Balance

3.2 平衡

Balance requires that every concept is bound with each position with equal probability. Concretely, we first calculate the entropy of all positions that a given concept c_n was bound to within dataset $\mathcal{D}$:

平衡要求每个概念与每个位置的绑定概率相等。具体而言，我们首先计算给定概念 c_n 在数据集 $\mathcal{D}$ 中绑定到的所有位置的熵：

$$\textbf{Balance}(c_n, \mathcal{D}) = Entropy\left[\psi^{-1}\left(\psi(\mathbf{F}^i, \mathbf{R}^i), c_n\right), i = 1, ..., |\mathcal{D}|\right], \qquad (4)$$

where the function *Entropy* computes the entropy of the distribution of the positions. Note that we constrain the computation to practical positions, i.e., excluding *null* during the calculation process. Then we aggregate by taking the expected value to obtain the balance over the entire dataset:

其中，函数 Entropy 计算位置分布的熵。注意，我们在计算过程中将计算限制在实际位置，即排除空值。然后，我们通过取期望值来聚合，以获得整个数据集的平衡：

$$\textbf{Balance}(\mathcal{D}) = \mathbb{E}\left[\textbf{Balance}(c_n, \mathcal{D})\right] = \sum_{c_n \in \mathcal{C}} \mathbb{P}(c_n)\textbf{Balance}(c_n, \mathcal{D}). \qquad (5)$$

This metric is upper bounded by $\log(M)$, corresponding to the entropy of a uniform distribution over $\mathcal{P}$. The lower the metric, the higher the skew of a dataset. We normalize by $\log(M)$ to obtain a value within $[0, 1]$.

该指标的上限为 $\log(M)$，对应于在 $\mathcal{P}$ 上的均匀分布的熵。指标越低，数据集的偏斜度越高。我们通过 $\log(M)$ 进行归一化，以获得 $[0, 1]$ 之间的数值。

So far we have defined completeness and balance for arbitrary sets of fillers and roles. Hereinafter, we will focus on binary relations, i.e. $|\mathbf{F}| = |\mathbf{R}| = 2$. Subscripts $_L$ and $_V$ denote the linguistic and visual perspectives under which a metric is computed.

到目前为止，我们已经为任意填充词和角色集定义了完整性和平衡性。在下文中，我们将重点关注二元关系，即 $|\mathbf{F}| = |\mathbf{R}| = 2$。下标 $_L$ 和 $_V$ 表示计算度量的语言学和视觉视角。

Practically, metrics are estimated by empirical counts. Next, we conduct experiments to demonstrate that generalization is hindered by incompleteness or imbalance under either perspective.

实际上，指标是通过经验计数来估计的。接下来，我们进行实验来证明，从任一角度来看，不完整或不平衡都会阻碍泛化。

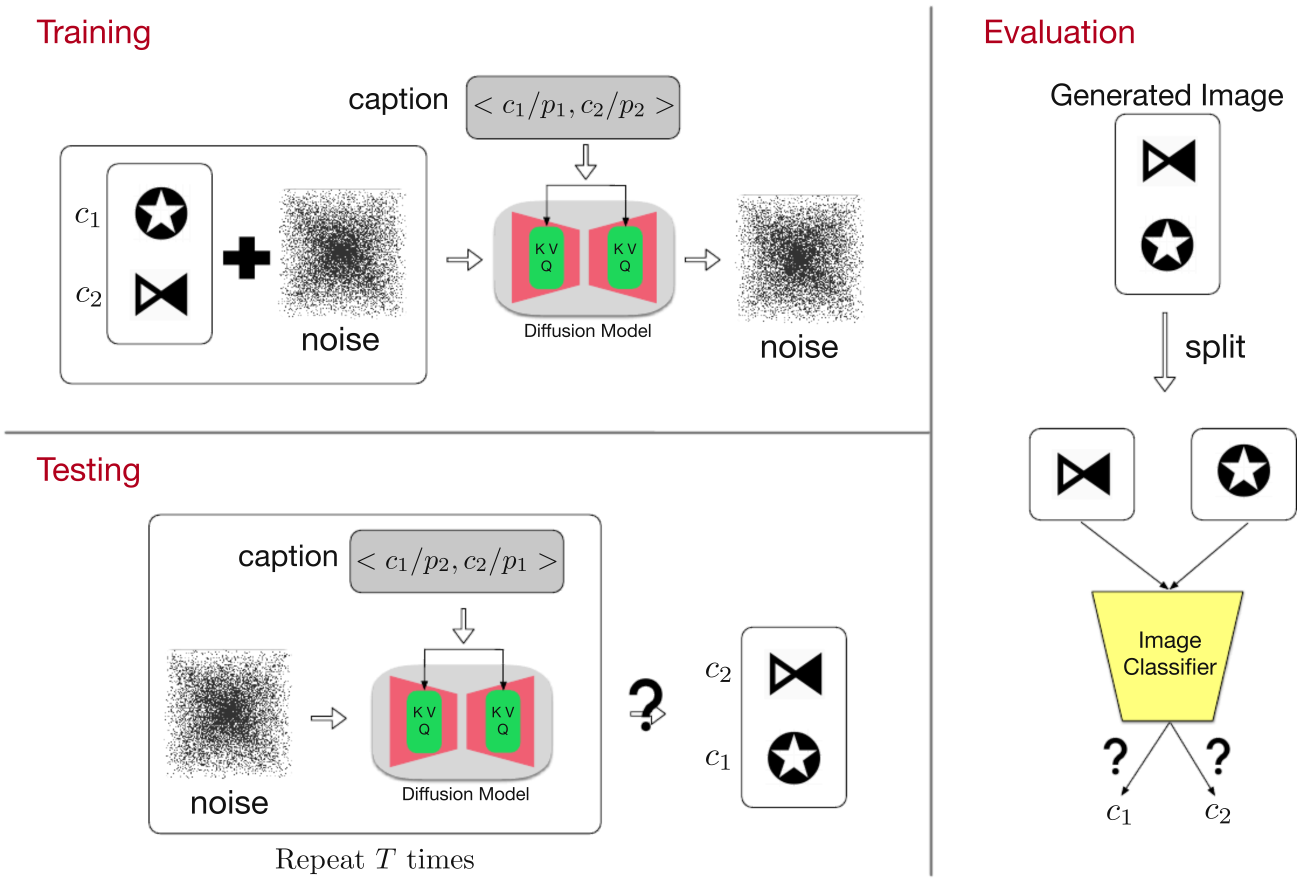


Fig. 4: Training, testing and evaluation pipeline. We train diffusion models to generate images of two concepts (c_1, c_2) with a specified spatial relation. Then the model is tested on unseen concept pairs to see whether the learned relations are generalizable.

图 4：训练、测试和评估流程。我们训练扩散模型来生成两个概念（c_1、c_2）的图像，并指定它们之间的空间关系。然后，在未见过的概念对上测试模型，以查看学习到的关系是否具有泛化性。

4 Experiments on Synthetic Images

合成图像的 4 个实验

4.1 Setup

4.1 设置

We use a set of unicode icons as concepts, and assign common nouns to them as their names. We vary the number of concepts, N , within $\{30, 40, 50, 60, 70, 80, 90\}$. We consider two symmetric binary relations: "on top of", "at the bottom of".

我们使用一组 Unicode 图标作为概念，并为它们分配常用名词作为名称。我们改变概念的数量，N，在 {30, 40, 50, 60, 70, 80, 90} 之间。我们考虑两个对称二元关系：“在上面”，“在下面”。

Images are constructed by drawing each icon on a 32x32 canvas, then stacking two icons vertically, resulting in 32x64 resolution. Captions are created from the template: "a(an) <icon_name> is <relation> a(an) <icon_name>".

图像通过在 32x32 画布上绘制每个图标，然后垂直堆叠两个图标，最终形成 32x64 分辨率来构建。标题从以下模板生成：“一个 是 一个 ”。

Training sets ($\mathcal{D}$) are sampled from $\mathcal{U}$ with systematic control for the four properties: $\textbf{Completeness}_L$, $\textbf{Completeness}_V$, $\textbf{Balance}_L$, $\textbf{Balance}_V$. To avoid confounders, we ensure the linguistic metrics are perfect when studying the effect of the visual metrics, and vice versa.

训练集（D）从 $\mathcal{U}$ 中采样，并对以下四个属性进行系统控制：$\textbf{Completeness}_L$，$\textbf{Completeness}_V$，$\textbf{Balance}_L$，$\textbf{Balance}_V$。为了避免混淆因素，我们在研究视觉度量指标的影响时确保语言度量指标完美，反之亦然。

Figure 3 illustrates training distribution with varied properties. We take the complementary set, $\mathcal{U} \setminus \mathcal{D}$ as the testing set. Testing instances are unseen in terms of the tuple representation (f_1, r_1, f_2, r_2) . Yet we show that perfect testing accuracy is possible when the training set properly supports the phenomena space, i.e. being complete and balanced.

图 3 展示了具有不同属性的训练分布。我们将互补集 U \ D 作为测试集。测试实例在元组表示 (f_1, r_1, f_2, r_2) 方面是未见过的。然而，我们证明了当训练集适当地支持现象空间，即完整且平衡时，可以实现完美的测试精度。

Our training, testing and evaluation pipeline is depicted in Figure 4. Following the architecture of [11], we train 350M UNet models on text-conditioned pixel-space diffusion, with T5-small [34] as the text encoder. The size of UNet is determined such that it is minimally above the threshold at which the model is capable of fully fitting the training set.

我们的训练、测试和评估流程如图 4 所示。遵循 [11], 的架构，我们在文本条件像素空间扩散上训练了 3.5 亿参数的 UNet 模型，使用 T5-small [34] 作为文本编码器。UNet 的大小确定为略高于模型能够完全拟合训练集的阈值。

This would avoid issues such as a lack of expressivity or under-training of an overparameterized model.[4] For evaluation, we compute accuracy of both icons being generated correctly[5]. The evaluation process is automated by pattern-matching icons with convolutional kernels.

这将避免诸如表达能力不足或过度参数化模型训练不足等问题。4 为了评估，我们计算了两个图标生成正确的准确率。5 评估过程通过将图标与卷积核进行模式匹配来自动化。

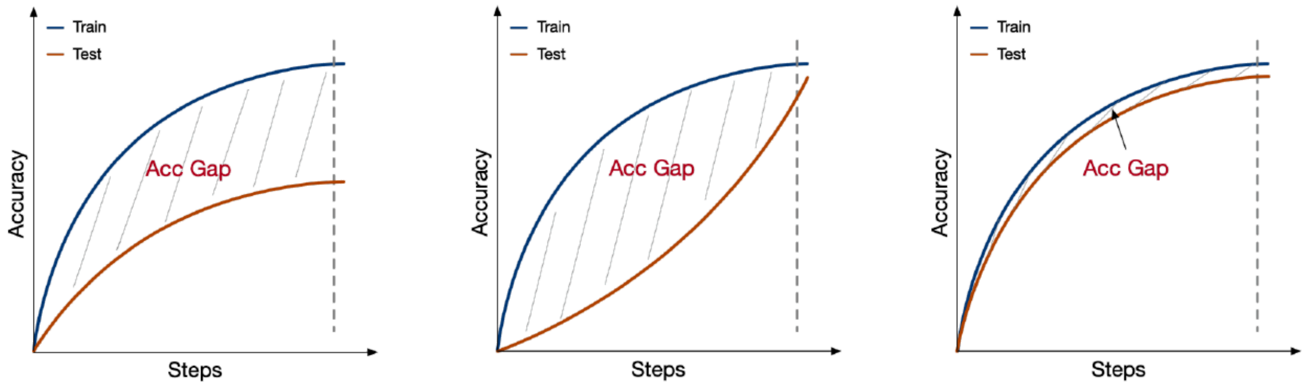


Fig. 5: Three types of learning dynamics are observed in our experiments. In the worst scenario (left), the testing accuracy plateaus and never converges. In the best scenario (right), the testing accuracy closely tracks the training accuracy until both converge to perfect. In the middle scenario (center), the testing accuracy climbs slower than the training accuracy, but is still able to converge to perfect at a delayed point after the training accuracy has already converged. In order to distinguish between the middle and best scenarios, we additionally report the accumulative gap between training and testing accuracy curves, which captures the timeliness of generalization.

图 5：在我们的实验中观察到三种类型的学习动态。在最坏情况下（左），测试精度趋于平稳，从未收敛。在最佳情况下（右），测试精度紧密跟踪训练精度，直到两者都收敛到完美。在中间情况下（中），测试精度比训练精度上升得慢，但仍然能够在训练精度已经收敛后延迟收敛到完美。为了区分中间情况和最佳情况，我们还报告了训练和测试精度曲线之间的累积差距，这反映了泛化的及时性。

Alongside the final testing accuracy, we report the accumulative difference between training and testing accuracy curves. This is because we have observed a third type of learning dynamics lying in between a generalization success and generalization failure (Figure 5), where the testing accuracy climbs slower than the training accuracy, but is still able to converge to perfect at a delayed point Only reporting on final testing accuracy would mask the qualitative difference that captures a notion of how timely generalization occurs.

除了最终测试精度，我们还报告了训练和测试精度曲线之间的累积差异。这是因为我们观察到了一种第三种学习动态，介于泛化成功和泛化失败之间（图 5)，其中测试精度比训练精度上升得慢，但仍然能够在延迟点收敛到完美）。仅报告最终测试精度会掩盖捕捉到泛化及时性的定性差异。

4.2 Results

4.2 结果

Visual incompleteness significantly impedes generalization. The last four experiments in Figure 6 (indexed by purple, brown, pink and grey) have low $\mathbf{Completeness}_V$. Their testing performance never reached 100%, even plateauing below 50% for smaller concept sets.

视觉上的不完整性极大地阻碍了泛化。图 6 中最后四个实验（用紫色、棕色、粉色和灰色标注）的 $\mathbf{Completeness}_V$ 很低。它们的测试性能从未达到 100%，甚至在较小的概念集中也停滞在 50% 以下。

Visual imbalance harms generalization when N is small. The first four experiments in Figure 6 (indexed by blue, orange, green and red) show the progression of increasing $\mathbf{Balance}_V$ while keeping full

Completeness$_V$. As **Balance**$_V$ grows, the testing accuracy consistently improves for all N . Increasing N can provide a remedy for a dataset with complete but imbalanced support.

当 N 很小时，视觉不平衡会损害泛化能力。图 6 中的前四个实验（分别用蓝色、橙色、绿色和红色表示）展示了在保持完整 **Completeness**$_V$ 的情况下，增加 **Balance**$_V$ 的过程。随着 **Balance**$_V$ 的增加，所有 N 的测试准确率都持续提高。增加 N 可以为具有完整但支持不平衡的数据集提供补救措施。

In contrast, larger N does not bring much help the support is incomplete.

相反，较大的 N 并没有带来太多帮助，支持不完整。

Linguistic incompleteness or imbalance harms generalization to a lesser degree, but they delay generalization. Figure 7 (left) shows that all cases achieve perfect or near-perfect testing accuracy, unless for the very small concept classes. This suggests that linguistic incompleteness and imbalance do not severely hinder whether the model is able to generalize ultimately. However, they do bring a negative effect by delaying the onset of generalization.

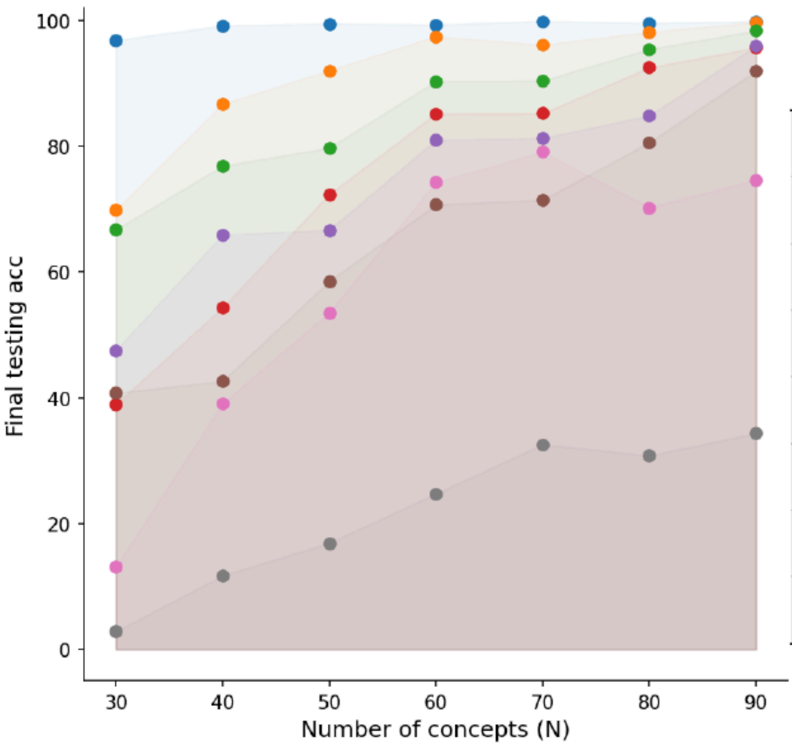
语言不完整或不平衡在一定程度上会损害泛化能力，但它们会延迟泛化。图 7（左）显示，除极小的概念类别外，所有情况都达到了完美或接近完美的测试精度。这表明语言不完整和不平衡不会严重阻碍模型最终是否能够泛化。然而，它们确实会通过延迟泛化的开始而带来负面影响。

This delay effect is revealed in Figure 7 (right). The takeaway is that, although the final testing accuracy is comparable, lack of $\textbf{Completeness}_L$ or $\textbf{Balance}_L$ causes the testing acc to largely lag behind training acc, which takes a longer time to catch up.

这种延迟效应在图7（右）中有所体现。结论是，尽管最终测试准确率相当，但缺乏$\textbf{Completeness}_L$ 或$\textbf{Balance}_L$ 会导致测试准确率大幅落后于训练准确率，需要更长时间才能追赶上来。

Similarly, we plot the generalization gap for the set of visual skew experiments in the appendix, observing the same trend. However, since both failing to generalize or having a prolonged duration before generalizing can lead to a large gap, this result has to be taken with a grain of salt.

类似地，我们在附录中绘制了视觉偏差实验集的泛化差距，观察到相同的趋势。然而，由于无法泛化或泛化前持续时间过长都会导致较大的差距，因此该结果需要谨慎对待。



	CPL(r1)	CPL(r2)	BLC	Coverage
● (blue)	100	100	100	20
● (orange)	100	100	84	55
● (green)	100	100	79	43
● (red)	100	100	71	50
● (purple)	84	84	79	70
● (brown)	75	75	68	56
● (pink)	100	50	68	50
● (grey)	50	50	0	25

Fig. 6: Under the definition of roles (r_1 , r_2) from **visual** perspective (“top", “bottom"), the plot (left) shows final testing accuracy against distributional properties of the training set. The legend and the corresponding metrics are summarized in the table (right). The results suggest that both **Completeness**$_V$ (CPL) and **Balance**$_V$ (BLC) are positively correlated with testing (generalization) performance. By contrast, a vanilla notion of data coverage is badly correlated with performance.

图 6：在角色定义（r_1 、r_2）（从视觉角度看，分别是“顶部”和“底部”）下，左侧的图显示了最终测试精度与训练集的分布属性之间的关系。图例和相应的指标在右侧的表格中进行了总结。结果表明，**Completeness**$_V$（CPL）和 **Balance**$_V$（BLC）都与测试（泛化）性能呈正相关。相比之下，数据覆盖率的简单概念与性能呈负相关。

Vanilla notions of coverage are a bad predictor for generalization. The rightmost columns in Figures 6 and 7 provide the training set’s coverage (%) over the universe $\mathcal{U}$. It can be seen that this does not correlate with the generalization performance.

香草覆盖率的概念是泛化性能的糟糕预测指标。图 6 和 7 中最右边的列显示了训练集在全集 $\mathcal{U}$ 上的覆盖率（%）。可以看出，这与泛化性能并不相关。

For example, the green run in Figure 6 outperformed red, purple, brown and pink runs, while having a much lower coverage than them.

Also noteworthy is that we intentionally select low-coverage datasets to demonstrate the fully-complete, fully-balanced case, which achieves perfect generalization for all N . This strongly indicates the problem of aiming for a generalizable model by recklessly scaling up coverage.

例如，图 6 中的绿色运行优于红色、紫色、棕色和粉色运行，而其覆盖率远低于它们。值得注意的是，我们有意选择低覆盖率数据集来展示完全完成、完全平衡的情况，该情况对所有 N 实现了完美的泛化。这强烈表明了通过鲁莽地扩大覆盖率来追求可泛化模型的问题。

In accordance with research on model bias, we argue that scaling up along the incorrect axes is dangerous because it may aggravate unintended bias, without necessarily benefiting generalization.

根据模型偏差研究，我们认为沿着错误轴进行扩展是危险的，因为它可能会加剧意外偏差，而不会 necessarily 提高泛化能力。

Increasing N eases generalization in all cases. The model consistently generalizes better when trained on more concepts, albeit to varying degrees. Figures 6 suggests that enlarging N is more helpful when the data is within a decent range of $\mathbf{Completeness}_V$ and $\mathbf{Balance}_V$ (e.g. 70-80).

增加 N 在所有情况下都能够提高泛化能力。模型在训练更多概念时始终表现出更好的泛化能力，尽管程度有所不同。图 6 表明，当数据在 $\mathbf{Completeness}_V$ 和 $\mathbf{Balance}_V$ 的合理范围内（例如 70-80）时，扩大 N 会更有帮助。

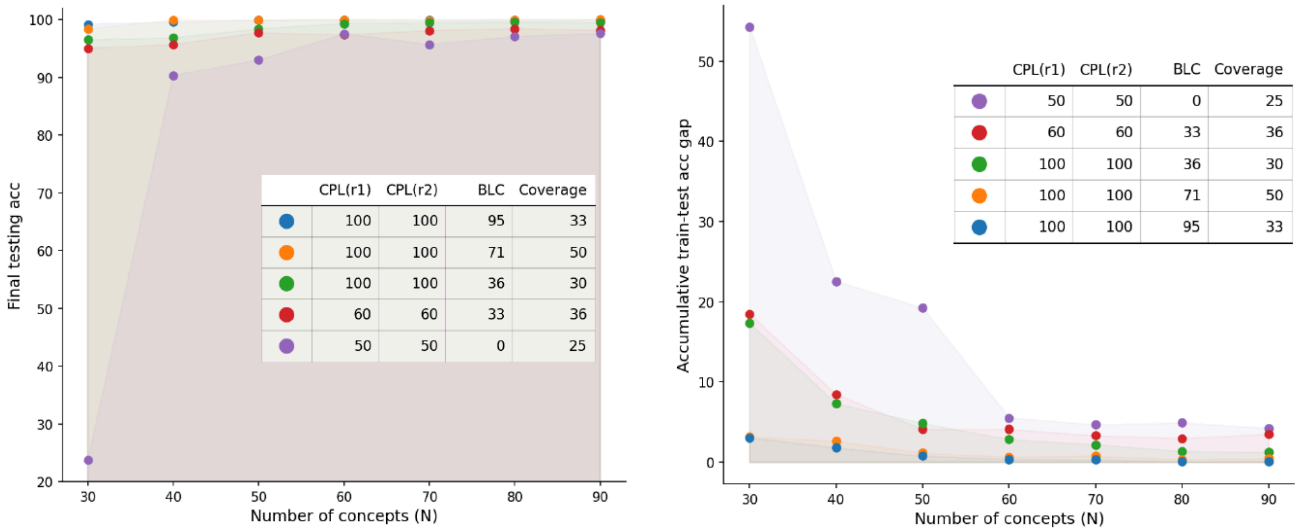


Fig. 7: Under the definition of roles (r_1 , r_2) , **linguistic**ally (“subj", “obj"), the plot shows final testing accuracy (left) and train-test accuracy gaps(right) against distributional properties of the training set. Legend and metrics are summarized in the table. The left plot suggests that linguistic incompleteness and imbalance harm generalization for small concept classes, while having less impact on the final testing accuracy for large concept classes. The right plot suggests that linguistic incompleteness and imbalance harm the timeliness of generalization, as indicated by larger train-test accuracy gaps.

图 7：在角色定义（r_1 、r_2）（语言学上分别为“主语”和“宾语”）下，该图展示了最终测试准确率（左）和训练-测试准确率差距（右）与训练集的分布特性之间的关系。图例和指标在表格中进行了总结。左图表明，对于小的概念类别，语言学上的不完整和不平衡会损害泛化能力，而对于大的概念类别，对最终测试准确率的影响较小。右图表明，语言学上的不完整和不平衡会损害泛化的及时性，如训练-测试准确率差距较大所示。

5 Experiments on Natural Images

自然图像的 5 个实验

5.1 Setup

5.1 设置

We extend our experiments to natural images using the What’sUp benchmark proposed by [16].[6] Our hypothesis is that higher Completeness and Balance of the training distribution lead to more generalizable learning outcomes. Each image in the What’sUp benchmark contains two objects, with a caption describing their spatial relations.

我们将实验扩展到使用 [16]6 提出的 What'sUp 基准的自然图像。我们的假设是，训练分布的更高完整性和平衡性会导致更具泛化性的学习结果。What'sUp 基准中的每张图像都包含两个物体，并带有描述它们空间关系的标题。

The spatial relations include two pairs of symmetric binary relations: *left/right of* and *in-front/behind*. Without loss of generality, we study *left/right of*, leaving the exploration of more than two relations for the future.

空间关系包括两对对称二元关系：左/右和前/后。不失一般性，我们研究左/右，将对超过两个关系的探索留待将来。

From an initial (complete and balanced) set of 308 samples with 15 unique concepts, subsamples are drawn where completeness and balance vary. For fair comparisons, the coverage of all subsamples is relatively the same. We train 470M pixel-space diffusion models on What'sUp image-caption pairs, with 64x32 resolution, 5e-4 learning rate and a batch size of 16.

从初始的（完整且平衡的）308 个样本集（包含 15 个独特的概念）中抽取子样本，子样本的完整性和平衡性各不相同。为了进行公平比较，所有子样本的覆盖率相对相同。我们在 What'sUp 图像-标题对上训练了 470M 像素空间扩散模型，分辨率为 64x32，学习率为 5e-4，批次大小为 16。

Early stopping is applied similarly to the synthetic setting. The length of training typically falls between 3000 and 6000 epochs. Hyperparameter tuning is described in the appendix.

早期停止与合成设置类似。训练的长度通常在 3000 到 6000 个 epoch 之间。超参数调整在附录中描述。

Since under the natural image setting the same object may occur at different image positions, evaluation could not be performed with predetermined pattern-matching kernels. We finetune ViT-B/16 [6] as an automatic evaluation engine to classify the object in the left or right crop of generated images.[7] A **Table 1:** What'sUp benchmark results varying **Completeness**$_V$ and **Balance**$_V$ **Table 2:** What'sUp benchmark results varying **Completeness**$_L$ and **Balance**$_L$

由于在自然图像设置下，同一物体可能出现在不同的图像位置，因此无法使用预先确定的模式匹配内核进行评估。我们微调了 ViT-B/16 [6] 作为自动评估引擎，以对生成图像的左侧或右侧裁剪中的物体进行分类。7 表 1：What'sUp 基准测试结果，变化 **Completeness**$_V$ 和 **Balance**$_V$ 表 2：What'sUp 基准测试结果，变化 **Completeness**$_L$ 和 **Balance**$_L$

Training Set Properties				Performance	
CPL(r_1)	CPL(r_2)	BLC	Cov.	Final Acc↑	Acc Gap ↓
100	50	63	47	18.75	84.55
80	73	77	50	19.52	73.87
87	87	75	49	25.93	68.41
100	100	73	44	10.17	80.49
100	100	88	49	28.50	64.89
100	100	100	48	48.64	33.67

Training Set Properties				Performance	
CPL(r_1)	CPL(r_2)	BLC	Cov.	Final Acc↑	Acc Gap ↓
50	100	63	47	57.14	40.72
80	73	77	50	60.00	39.83
87	87	75	49	62.04	38.44
100	100	73	44	62.71	33.90
100	100	80	37	65.04	30.21
100	100	88	49	67.29	32.90

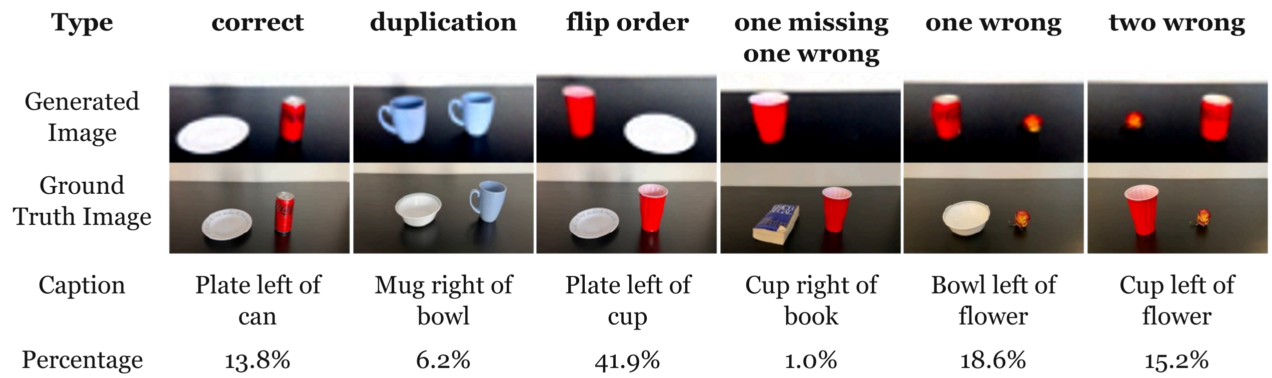


Fig. 8: Qualitative examples of generated images and the corresponding ground truth images, including a categorization of failure types and their frequencies.

图 8：生成的图像和相应的真实图像的定性示例，包括故障类型分类及其频率。

"blank" label is added to the classifier in order to indicate when a model fails to generate an object. Importantly, a secondary result of our work is demonstrating shortcomings of the widely used evaluation methods for image synthesis, such as CLIPScore [10], VQA with VLMs [13,26], bounding-box evaluation with detectors [8, 13].

为了表明模型何时无法生成对象，在分类器中添加了"空白"标签。重要的是，我们工作的另一个结果是证明了广泛使用的图像合成评估方法的不足，例如 CLIPScore [10],、使用 VLM 的 VQA [13,26]、使用检测器的边界框评估 [8, 13]。

See Section 6 for discussions of when they fall short.

参见第 6 节，讨论它们何时不足。

5.2 Results

5.2 结果

On natural images, it is much harder to generalize, probably because the objects' absolute position and postures can vary even when the relative spatial positions are determined. Another likely cause is the small number of concepts, as suggested by Section 4.2 that generalization tends to plateau at 50% for a small concept set.

在自然图像上，泛化要困难得多，这可能是因为即使相对空间位置确定了，物体的绝对位置和姿态也会发生变化。另一个可能的原因是概念数量少，正如第 4.2 节所述，对于小型概念集，泛化往往在 50% 处趋于平稳。

Nevertheless, the relative performance across different training set properties still conveys a meaningful message. Table 1 shows a consistent trend of generalization performance influenced by $\mathbf{Completeness}_V$ and $\mathbf{Balance}_V$. The final accuracy gets higher and the generalization gap gets lower as the training distribution moves towards being fully complete and balanced.

然而，不同训练集属性的相对性能仍然传达了一个有意义的信息。表 1 显示了由 $\mathbf{Completeness}_V$ 和 $\mathbf{Balance}_V$ 影响的泛化性能的一致趋势。随着训练分布向完全完整和平衡的方向移动，最终的准确率会更高，泛化差距会更小。

$\mathbf{Completeness}_L$ and $\mathbf{Balance}_L$ achieve a similar effect, as shown by Table 2.

$\mathbf{Completeness}_L$ 和 $\mathbf{Balance}_L$ 实现了类似的效果，如表 2 所示。

As with our previous findings, visual skew imposes a more severe challenge, compared with the same amount of distributional skew on the linguistic side. Our explanation is that, since the network is modeling the pixel space, it is more directly affected by the skew of the observed pixel-space distribution, while skew of the language-space distribution impacts the result rather indirectly.

与我们之前的发现一致，视觉倾斜比语言侧相同程度的分布倾斜带来了更严峻的挑战。我们的解释是，由于网络对像素空间进行建模，因此它受到观察到的像素空间分布倾斜的直接影响，而语言空间分布的倾斜则对结果的影响较为间接。

Upon closer examination of model outputs, the most common error was generating the correct objects with flipped order. This suggests that mapping fillers across domains is easy, but learning to map roles when only observing role-filler bindings is hard. Other errors include generating a blank or duplicating the object.

仔细检查模型输出后，最常见的错误是生成顺序颠倒的正确对象。这表明跨域映射填充词很容易，但仅通过观察角色-填充词绑定来学习映射角色却很难。其他错误包括生成空白或重复对象。

Figure 8 visualizes examples of correct and incorrect generations.

图 8 展示了正确和错误生成的示例。

6 Discussions

6 个讨论

We present a conceptual framework and formal metrics to study the contributing factors of generating images with correct spatial relations. Clearly, our work triggers many open questions that are worth future exploration. Appendix I further discusses limitations of this work and future directions.

我们提出了一个概念框架和正式指标来研究生成具有正确空间关系的图像的贡献因素。显然，我们的工作引发了许多值得未来探索的开放性问题。附录 I 进一步讨论了这项工作的局限性和未来的方向。

Are spatial relations distinguishable within unimodal spaces? We have mainly focused on what enables entity-relation compositions to be successfully conveyed from the text to the image domain. However, this question is meaningful only under the assumption that different roles and fillers are distinctly encoded in unimodal spaces.

在单模空间内，空间关系是否可以区分？我们主要关注的是什么使得实体-关系组合能够成功地从文本域传递到图像域。然而，这个问题只有在不同角色和填充物在单模空间中被明确编码的假设下才有意义。

We have evidence that, perhaps surprisingly, this assumption does not always hold in existing approaches.

我们有证据表明，也许令人惊讶的是，这种假设并不总是适用于现有的方法。

We train probing classifiers to extract the positional role of nouns from text encodings. More details are available in Appendix E. We find that probes trained with the CLIP text encoder can only overfit the training data, but not generalizing. This indicates the inherent weakness of CLIP text encoder to provide consistent signals of spatial positions.

我们训练探测分类器来从文本编码中提取名词的位置角色。更多细节见附录 E。我们发现用 CLIP 文本编码器训练的探测器只能过拟合训练数据，而不能泛化。这表明 CLIP 文本编码器在提供一致的空间位置信号方面存在固有弱点。

This finding aligns with existing criticism on CLIP essentially being a bag-of-word model [48]. By contrast, probing experiments with T5 [34] encoder and the encoder of a pretrained vision-language model [9] succeed with near-perfect generalization. This makes T5 and VLMs naturally better candidates for training text-to-image models.

这一发现与现有的批评一致，即 CLIP 本质上是一个词袋模型 [48]。相比之下，对 T5 [34] 编码器和预训练视觉语言模型 [9] 编码器的探测实验取得了近乎完美的泛化效果。这使得 T5 和 VLM 成为训练文本到图像模型的自然更好的候选者。

On the image side, the capacity to represent positions can be theoretically guaranteed by image-patch positional encodings, readily compatible with attention blocks in the diffusion architecture. However, to our surprise, most of the open-source diffusion implementations [31] omit this step. We noticed this issue as our initial experiments failed unexpectedly.

在图像侧，图像块位置编码可以理论上保证表示位置的能力，并且与扩散架构中的注意力块兼容。然而，令我们惊讶的是，大多数开源扩散实现 [31] 省略了这一步。我们在最初的实验中意外失败时注意到了这个问题。

The problem was fixed after we modified the architecture to include image positional encodings. We posit that the lack of image positional encodings imposes a representational deficiency[8] leading to heavy reliance on pixel correlations and unexpected testing behavior. Ablation studies in Appendix F support this argument.

在我们将架构修改为包含图像位置编码后，问题得到了解决。我们认为，缺乏图像位置编码会导致表示能力不足[NT]8[NT]，从而导致对像素相关性的过度依赖和意外的测试行为。附录 F 中的消融研究支持这一论点。

In short, we point out the importance of a text encoder that distinguishes positional roles and an image decoder that has the representational power for spatial information. We emphasize that these are not only important preconditions for our main analysis presented in this paper, but should also be crucial considerations in future generative models or models that do spatial reasoning.

简而言之，我们指出了区分位置角色的文本编码器和具有空间信息表示能力的图像解码器的重要性。我们强调，这些不仅是我们本文中主要分析的重要先决条件，而且也

应该是未来生成模型或进行空间推理模型的关键考虑因素。

Generation in the Latent Space There are abundant studies on latent generation methods [4, 30, 36, 37] that are able to achieve higher resolution. However, the progress towards spatially consistent high-resolution synthesis might be hampered when the data coverage of underlying phenomena does not proportionally scale with resolution.

潜在空间中的生成 在潜在生成方法方面存在大量研究 [4, 30, 36, 37]，这些方法能够实现更高的分辨率。然而，当潜在现象的数据覆盖范围与分辨率不成比例地扩展时，朝着空间一致的高分辨率合成的进展可能会受到阻碍。

To test whether our arguments carry over to latent-space generation, we conduct experiments with a pretrained (frozen) VAE (stabilityai/stable-diffusion-2-1). We increase the input resolution by a factor of four because VAE applies compression. The results (Appendix G) match our intuition that a latent space does not affect the validity of our conceptual and formal frameworks.

为了检验我们的论点是否适用于潜在空间生成，我们使用预训练（冻结）VAE（stabilityai/stable-diffusion-2-1）进行了实验。由于VAE应用了压缩，我们通过将输入分辨率提高四倍来进行实验。结果（附录G）符合我们的直觉，即潜在空间不会影响我们概念和形式框架的有效性。

CPL and BLC consistently correlate with generalization performance. Also in accordance with our existing findings, linguistic skew harms final testing performance to a lesser degree, but delays generalization, as evidenced by large acc gaps. Finally, the latent diffusion results again verified that larger coverage cannot compensate for a poor CPL or BLC.

CPL 和 BLC 与泛化性能始终保持相关性。同样根据我们现有的发现，语言偏差对最终测试性能的影响较小，但会延迟泛化，如较大的 acc 差距所表明的那样。最后，潜在扩散结果再次验证了更大的覆盖范围无法弥补较差的 CPL 或 BLC。

Two factors may explain the similarity in results between pixel and latent spaces. First, the latent space feature maps have spatial correspondence with the image. Second, the VAE [17] does not have a language component, as such the language-to-vision communication channel is captured by diffusion.

两个因素可以解释像素空间和潜在空间结果的相似性。首先，潜在空间特征图与图像具有空间对应关系。其次，VAE[17]没有语言成分，因此语言到视觉的通信通道由扩散模型捕获。

While better features may be provided by VAE, they lack any cross-modal correspondence. In summary, increasing the phenomenological coverage and increasing resolution are both important. We leave the questions open on extending our formal notions to superresolution models as well as more nuanced relations.

虽然 VAE 可以提供更好的特征，但它们缺乏跨模态对应关系。总之，增加现象学覆盖范围和提高分辨率都很重要。我们将扩展我们形式化概念到超分辨率模型以及更细致关系的问题留待进一步研究。

Text-to-Image Evaluation Methods We rely on several heuristics when automating the evaluation of generated images. This was feasible only when 1) objects are center aligned, and 2) the background is clean. Ultimately, we are interested in evaluating relations with cluttered scenes and with greater appearance variability.

文本到图像评估方法 我们在自动评估生成图像时依赖于一些启发式方法。这只有在以下情况下才可行：1）对象居中对齐，以及 2）背景干净。最终，我们感兴趣的是评估具有杂乱场景和更大外观变化的关系。

Besides finetuning a ViT classifier, we have explored existing off-the-shelf models, but found them limited in one way or another. CLIPScore is known to pay less attention to token orders. Indeed, CLIPScore judges correctly in only 37% of the times in our setting.

除了微调 ViT 分类器，我们还探索了现有的现成模型，但发现它们在某种程度上受到限制。CLIPScore 众所周知对词元顺序的关注较少。事实上，在我们的设置中，CLIPScore 只有 37% 的时间能够正确判断。

Evaluating spatial relations by comparing bounding box locations produced by detectors offers a more structured approach, yet it is restricted to the object classes available in the detection pretraining. For example, “headphones" and “tape" are classes in the What’sUp benchmark that do not belong to any of the popular detection datasets, rendering most of the detection models inappropriate for our experiments.

通过比较检测器生成的边界框位置来评估空间关系提供了一种更结构化的方法，但它仅限于检测预训练中可用的目标类别。例如，“耳机”和“磁带”是 What’sUp 基准测试中的类别，不属于任何流行的检测数据集，这使得大多数检测模型不适合我们的实验。

Open-vocabulary detectors [27] circumvents this problem. But practical issues still exist, such as redundancy in the predicted bounding boxes and sensitivity to text prompts.

开放词汇检测器 [27] 规避了这个问题。但实际问题仍然存在，例如预测边界框的冗余和对文本提示的敏感性。

Aside from using CLIP or detectors, the literature has also suggested using vision-language foundation models (VLMs) for synthesized images evaluation. In addition to the apparent drawback of slow inference, it remains questionable whether VLMs comprehend relations in the first place [48].

除了使用 CLIP 或检测器，文献还建议使用视觉语言基础模型 (VLMs) 来评估合成图像。除了明显的推理速度慢的缺点外，VLMs 是否能够理解关系本身仍然值得怀疑 [48]。

Our attempts at LLaVA- or BLIP2-VQA were unsuccessful for multiple reasons such as inability to recognize spatial relation (Table A3), and unbalanced precisions across object classes (Table A5). See Appendix H for more analysis. We hope our investigation calls into question the effectiveness of existing metrics on spatial consistency, which might inadvertently mask the weakness of text-to-image models.

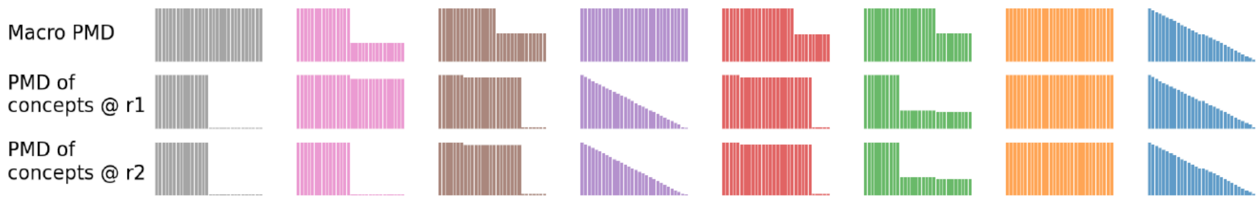
我们在尝试使用 LLaVA 或 BLIP2-VQA 时遇到了多个问题，例如无法识别空间关系（表 A3)）以及不同物体类别之间精度不平衡（表 A5）。更多分析请参见附录 H。我们希望我们的研究能够引起人们对现有空间一致性指标有效性的质疑，这些指标可能无意中掩盖了文本到图像模型的弱点。

Other data distributional properties Our metrics are designed around completeness and balance, but they may not capture other distributional properties, such as the Zipfian-Uniform axis. Figure 9 plots the probability mass distribution of datasets we used in Section 4. We see three shapes: uniform, 2-stages, and “wedge". This provides new insights: 1) Macro-PMD may hide skew, visible when we plot role-specific-PMD (row 2&3). E.g. grey and purple instances (col 1&4) exhibit uniform macro-PMD, but biased role-specific-PMD – correlated to poor generalization. 2) A non-uniform PMD may not indicate poor generalization. E.g. the blue instance (col 8) has non-uniform PMD, yet achieves great performance – perfect CPL and BLC scores.

其他数据分布特性 我们的指标围绕完整性和平衡性设计，但它们可能无法捕捉其他分布特性，例如齐夫-均匀轴。图 9 绘制了我们在第 4 节中使用的数据集的概率质量分布。我们看到三种形状：均匀、两阶段和"楔形"。这提供了新的见解：1）宏观 PMD 可能隐藏了偏斜，当我们绘制角色特定 PMD（第 2 行和第 3 行）时可见。例如，灰色和紫色实例（第 1 列和第 4 列）表现出均匀的宏观 PMD，但偏置的角色特定 PMD - 与泛化能力差相关。2）非均匀 PMD 可能不代表泛化能力差。例如，蓝色实例（第 8 列）具有非均匀 PMD，但实现了出色的性能 - 完美的 CPL 和 BLC 分数。

Future work may explore other data distributional properties and their correlation with generalization.

未来的工作可以探索其他数据分布特性及其与泛化的相关性。



CPL(r1)	50	100	75	97	84	100	100	100
CPL(r2)	50	0	75	97	84	100	100	100
BLC	0	68	68	71	79	84	100	100
Testing acc (pixel) ↑	2.79	13.1	43.14	38.97	47.55	69.87	96.88	98.93
Acc gap (pixel) ↓	88.77	81.48	65.29	64.56	62.35	41.23	10.65	12.27
Testing acc (latent) ↑	2.87	15.75	45.66	32.18	45.47	61.54	96.01	99.88
Acc gap (latent) ↓	92.61	79.45	60.39	70.82	59.95	45.52	10.37	5.6

Fig. 9: Probability mass distribution (PMD) of various training sets used in Section 4. Testing performance is correlated more with CPL and BLC, than with the uniformity of PMD. Also note that the PMD of concepts at individual roles may not be uniform although it appears to be uniform on the macro level.

图 9：第 4 节中使用的各种训练集的概率质量分布 (PMD)。测试性能与 CPL 和 BLC 的相关性高于与 PMD 的均匀性。此外，请注意，尽管在宏观层面上 PMD 似乎是均匀的，但单个角色的概念的 PMD 可能并不均匀。

7 Conclusion

7 结论

Text-to-Image synthesis, despite recent breakthroughs in fidelity, appearance diversity and texture granularity, still struggles with relations. As the community strives for larger datasets to better cover the natural distribution, there is a lack of study on the axes along which phenomenological coverage can be meaningfully enlarged.

尽管文本到图像合成在保真度、外观多样性和纹理粒度方面取得了最近的突破，但它仍然难以处理关系。随着社区努力构建更大的数据集以更好地覆盖自然分布，人们缺乏对可以有效扩大现象学覆盖范围的轴线的研究。

This work presents the first effort to formally characterize training coverage, in the context of learning spatial relations. We introduce completeness and balance metrics under both the linguistic and visual perspectives. Our experiments on synthetic and natural data consistently suggest that models trained on more complete and balanced datasets have greater generalization potential.

本研究首次尝试正式描述空间关系学习中的训练覆盖率。我们从语言和视觉两个角度引入了完整性和平衡性指标。我们在合成数据和真实数据上的实验一致表明，在更完整和平衡的数据集上训练的模型具有更高的泛化潜力。

We see this work as a stepping stone towards text-to-image models that can faithfully generate relations in general, including implicit relations entailed by verbs. We hope to inspire research on structured image evaluation, architectures for modeling role-filler bindings and formal frameworks of generalization.

我们将这项工作视为通向文本到图像模型的垫脚石，该模型能够忠实地生成一般关系，包括动词隐含的关系。我们希望激发对结构化图像评估、角色填充绑定建模架构和泛化形式框架的研究。

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Appendix

附录

A Completeness and Balance Scores of Data in the Wild

野外数据完整性和平衡性评分

Table A1: We parse three large-scale image-caption datasets, namely VGR, Nocaps, and Flickr30K. Then we compute **Completeness** and **Balance** on each corresponding subset involving spatial relations. LR stands for left-right, TB stands for top-bottom, and FB stands for front-behind. "-complete" means a subsample that is both linguistically and visually complete according to our definitions.

表 A1：我们解析了三个大规模图像-标题数据集，分别是 VGR、Nocaps 和 Flickr30K。然后，我们计算了每个包含空间关系的子集的完整性和平衡性。LR 代表左右，TB 代表上下，FB 代表前后。"-complete"表示根据我们的定义在语言和视觉上都完整的子样本。

	Dataset	#pairs	#unique images	#unique captions	#unique concepts	CPL_V	CPL_L	BLC_V	BLC_L
		Basic Statistics				**Skew Metrics**			
VGR	Original	21,944	—	—	—	—	—	—	—
	Spatial	19,419	4,348	12,163	885	—	—	—	—
	LR	15,482	3,244	9,816	695	80	100	93	99
	TB	2,505	1,553	1,513	439	66	73	49	62
	FB	11,162	527	834	209	69	95	58	98
	LR-complete	14,352	3,074	8,766	414	100	100	96	99
	TB-complete	784	488	395	77	100	100	57	69
	FB-complete	741	335	449	61	100	100	68	100
Nocaps	Original	45,000	—	—	—	—	—	—	—
	Spatial	14,583	3,827	14,241	2,975	—	—	—	—
	TB	12,726	3,540	12,714	2,273	63	62	35	31
	FB	1,854	1,051	1,854	703	65	64	57	45
	TB-complete	4,922	2,176	4,921	369	100	100	46	48
	FB-complete	718	491	718	110	100	100	75	62
Flickr30k	Original	155,070	—	—	—	—	—	—	—
	Spatial	44,918	20,784	43,419	4,205	—	—	—	—
	TB	37,541	18,436	37,462	3,003	67	66	36	29
	FB	7,351	5,333	7,350	1,483	65	63	57	38
	TB-complete	30,162	16,132	30,102	872	100	100	38	31
	FB-complete	4,016	3,185	4,016	265	100	100	66	48

On a high level, we seek to identify the causes for the dissatisfying relation learning in text-to-image generation from the data distribution angle. Our theory shows promising predictivity both on synthetic images and on real images captured in a clean and controlled setting.

从数据分布的角度，我们旨在从高层次上识别文本到图像生成中令人不满意关系学习的原因。我们的理论在合成图像和在干净受控环境中捕获的真实图像上都显示出有希望的预测能力。

Ultimately, we expect to extend our investigation to real images *in the wild*. To facilitate research along this vein, we have parsed and analyzed three large-scale image-caption datasets, namely VGR (Visual

Genome Relation) [48], Nocaps (Novel Object Captioning) [1], and Flickr30K [46].

最终，我们希望将我们的研究扩展到现实世界中的真实图像。为了促进这方面的研究，我们解析并分析了三个大型的图像-字幕数据集，分别是 VGR（Visual Genome Relation）[48], Nocaps（Novel Object Captioning）[1], 和 Flickr30K [46]。

We draw subsamples by matching spatial phrases in the captions for three spatial relations: left-right (LR), top-bottom (TB) and front-behind (FB). We present **Completeness** and **Balance** scores, along with basic statistics for each of these subsamples in Table A1.

我们通过匹配标题中的空间短语来抽取子样本，用于三种空间关系：左右 (LR)、上下 (TB) 和前后 (FB)。我们在表 A1 中展示了完整性和平衡度分数，以及每个子样本的基本统计数据。

VGR is carefully curated from Visual Genome [18] for the study of relational understanding in vision-language models. Thus, most of the examples in VGR involve a spatial relation. On the other hand, only a small subset in Nocaps or Flick30k contains spatial information.

VGR 是从 Visual Genome [18] 中精心挑选出来的，用于研究视觉语言模型中的关系理解。因此，VGR 中的大多数示例都涉及空间关系。另一方面，Nocaps 或 Flick30k 中只有很小一部分包含空间信息。

The supportive data size further shrinks if we constrain our subsamples to be complete under our definition[9]. As can be seen from highlighted rows in Table A1, **the largest complete subsample only constitutes a tiny fraction of the original data**, plus being highly unbalanced.

如果我们将子样本限制为根据我们的定义9完整的，则支持数据的规模将进一步缩小。如表A1中突出显示的行所示，最大的完整子样本仅占原始数据的极小部分，并且高度不平衡。

This emphasizes the importance of designing architectures or algorithms to enable generalization even from skewed data sources.

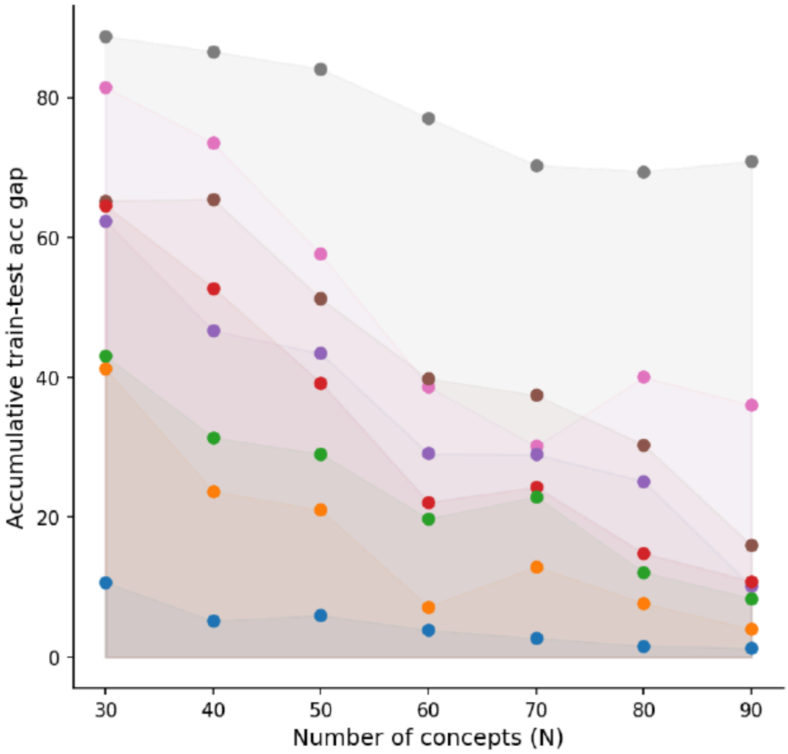
这强调了设计架构或算法以从偏斜数据源中实现泛化的重要性。

B Experiments on Synthetic Data — Additional Results

B. 合成数据实验 - 附加结果

Figure A1 plots for the train-test accuracy gap under varying $\textbf{Completeness}_V$ and $\textbf{Balance}_V$ scores. Moreover, detailed learning curves are comprehensively presented in Figure A2 and A3.

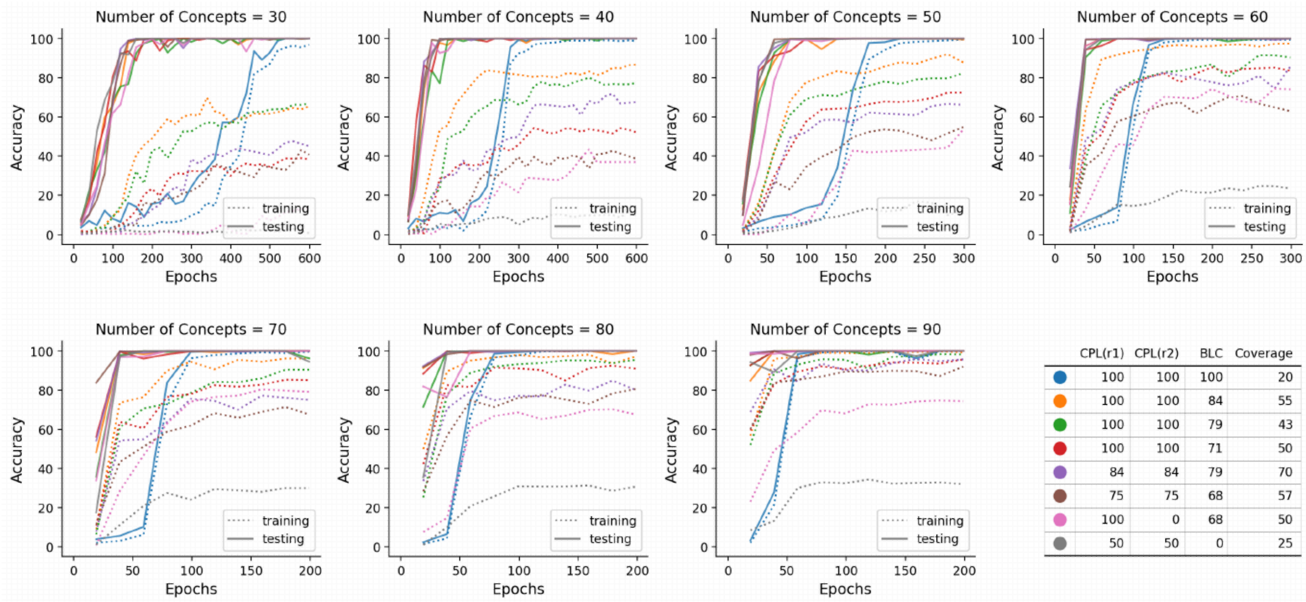
图A1绘制了在不同$\textbf{Completeness}_V$和$\textbf{Balance}_V$分数下训练-测试准确率差距。此外，图A2和A3全面展示了详细的学习曲线。



	CPL(r1)	CPL(r2)	BLC	Coverage
●	50	50	0	25
●	100	50	68	50
●	75	75	68	56
●	84	84	79	70
●	100	100	71	50
●	100	100	79	43
●	100	100	84	55
●	100	100	100	20

Fig. A1: Under the definition of roles (r_1 , r_2) from **visual** perspective (“top", “bottom"), the plot shows final testing accuracy against distributional properties of the training set. Legend and corresponding metrics are summarized in the table. This plot suggests that visual incompleteness and imbalance lead to larger gaps because both the final generalization performance and generalization speed are hampered.

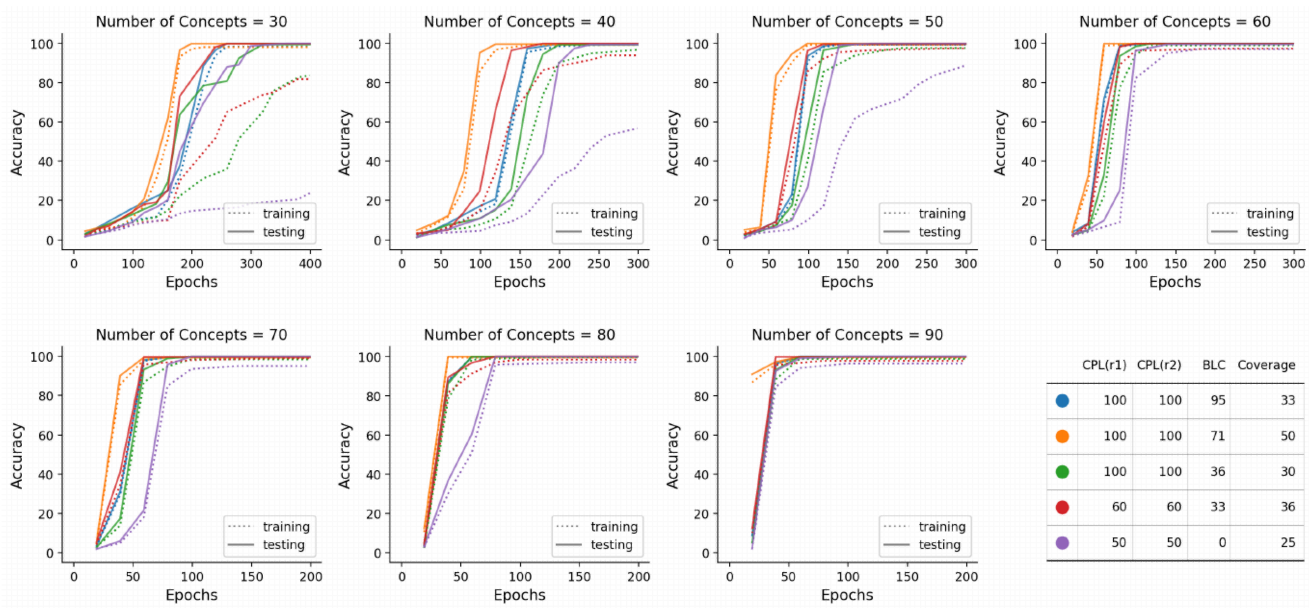
图 A1：在角色定义（r_1 、r_2）（从视觉角度看，分别是“顶部”和“底部”）下，该图显示了最终测试精度与训练集的分布属性的关系。图例和相应的指标在表格中进行了总结。该图表明，视觉上的不完整性和不平衡会导致更大的差距，因为最终的泛化性能和泛化速度都会受到影响。



	CPL(r1)	CPL(r2)	BLC	Coverage
●	100	100	100	20
●	100	100	84	55
●	100	100	79	43
●	100	100	71	50
●	84	84	79	70
●	75	75	68	57
●	100	0	68	50
●	50	50	0	25

Fig. A2: Detailed results on the impacts of $\mathbf{Completeness}_V$ and $\mathbf{Balance}_V$. Panels are separated according to the number of concepts. Training accuracy always converges to perfect, whereas testing accuracy is hampered to varying degrees by visual incompleteness or imbalance. This degradation is more severe for a small number of concepts.

图 A2: $\mathbf{Completeness}_V$ 和 $\mathbf{Balance}_V$ 影响的详细结果。面板根据概念数量进行划分。训练精度始终收敛到完美，而测试精度因视觉不完整或不平衡而受到不同程度的阻碍。这种下降对于少量概念更为严重。



	CPL(r1)	CPL(r2)	BLC	Coverage
●	100	100	95	33
●	100	100	71	50
●	100	100	36	30
●	60	60	33	36
●	50	50	0	25

Fig. A3: Detailed results on the impacts of $\mathbf{Completeness}_L$ and $\mathbf{Balance}_L$. Panels are separated according to the number of concepts. Training accuracy always converges to perfect, whereas testing accuracy is hampered to varying degrees by linguistic incompleteness or imbalance. This degradation is more severe for a small number of concepts.

图 A3: $\textbf{Completeness}_L$ 和 $\textbf{Balance}_L$ 影响的详细结果。图示根据概念数量进行划分。训练准确率始终收敛到完美，而测试准确率则受到语言不完整或不平衡的程度不同影响。这种下降在概念数量较少时更为严重。

C Experiments on Synthetic Data — Details

合成数据上的 C 实验 - 细节

C.1 Icons and Their Names

C.1 图标及其名称

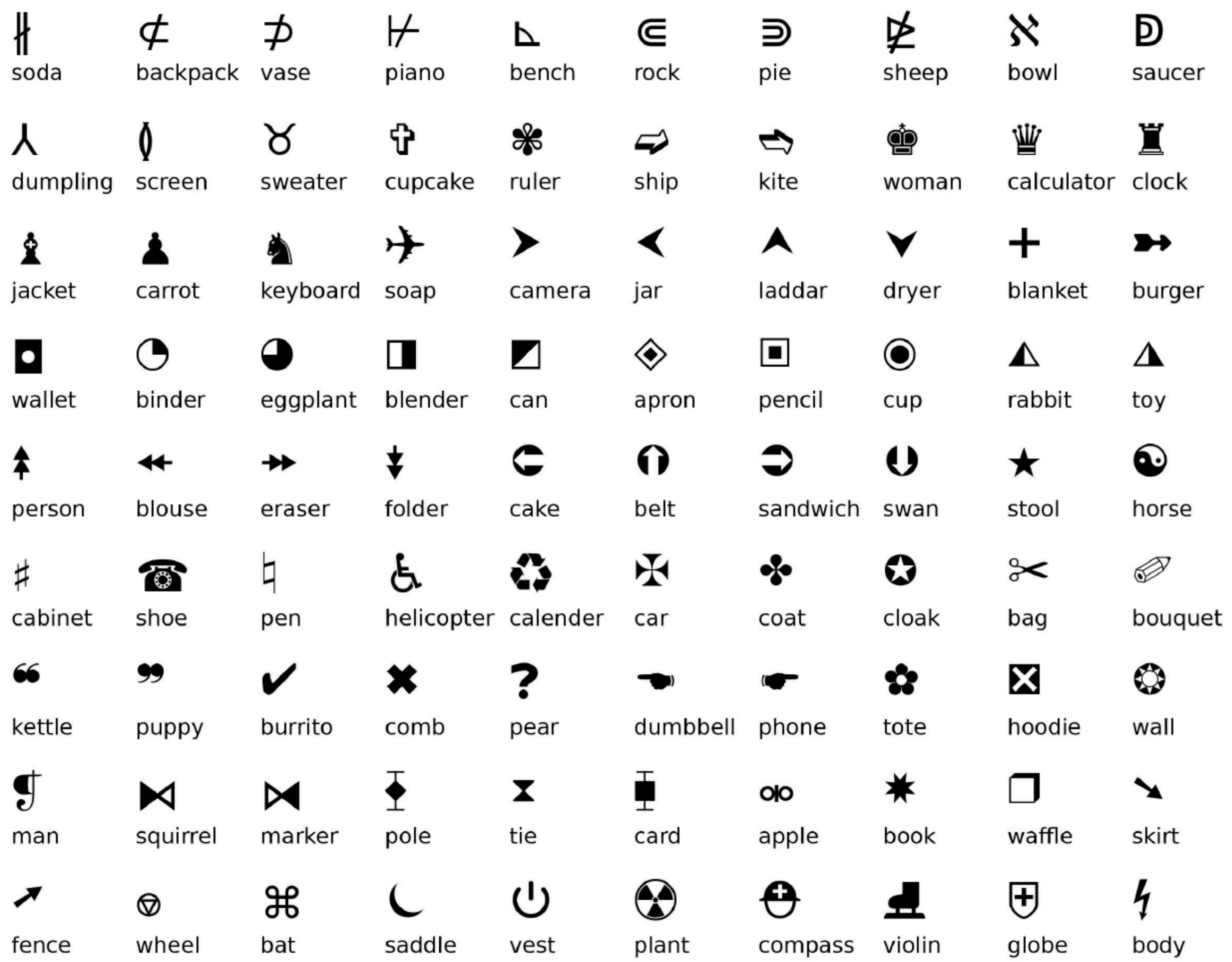


Fig. A4: 90 icons paired with 90 common nouns as their names.

图 A4：90 个图标与 90 个常用名词配对。

We consider unicode icons as concepts, each paired with a concrete noun as the identity. To create an image, we draw icons[10] on a white canvas. Figure A4 shows 90 individual icons which we leverage in our synthetic experiments.

我们将 Unicode 图标视为概念，每个概念都与一个具体的名称配对作为标识。为了创建图像，我们在白色画布上绘制图标10。图 A4 显示了 90 个单独的图标，我们在合成实验中利用这些图标。

C.2 Model Configs

C.2 模型配置

```
layers_per_block = 2
block_out_channels = (64, 256, 1024)
down_block_types = (
    "DownBlock2D",
    "CrossAttnDownBlock2D",
    "CrossAttnDownBlock2D",
)
up_block_types = (
    "CrossAttnUpBlock2D",
```

```
        "CrossAttnUpBlock2D",
        "UpBlock2D",
    )
    cross_attention_dim = 128
    patch_size = 2
    conv_in_kernel = 3
    conv_out_kernel = 3
```

C.3 Training Configs

C.3 训练配置

We first pretrain the model on single-icon crops to familarize it with the appearance of each icon. Then we finetune the model to generate two icons with different training set statistics.

我们首先在单个图标裁剪上预训练模型，使其熟悉每个图标的外观。然后，我们微调模型以生成两个具有不同训练集统计信息的图标。

Pretraining:

预训练：

```
image_size = (32,32)
```

```
mixed_precision = "fp16"
```

```
noise_schedule = "squaredcos_cap_v2"
```

```
learning_rate = 1e-4
```

```
lr_warmup_steps = 1000
```

```
train_batch_size = 16
```

num_train_timesteps = 100 # d i f f u s i o n s t e p s

num_train_timesteps = 100 # diffusion steps

Finetuning:

微调：

```
image_size = (64,32) # H, W
```

```
mixed_precision = "fp16"
```

noise_schedule = "squaredcos_cap_v2"

noise_schedule = "squaredcos_cap_v2"

```
learning_rate = 1e-4
```

```
lr_warmup_steps = 1000
```

```
train_batch_size = 16
```

num_train_timesteps = 100 # diffusion steps

num_train_timesteps = 100 # diffusion steps

C.4 Hyperparameter Search

C.4 超参数搜索

layers_per_block $\in$ $\{2, 3, 4\}$.

layers_per_block $\in$ $\{2, 3, 4\}$。

block_out_channels $\in$ $\{(32, 128, 512), (128, 256, 512), (64, 256, 1024), (64, 512, 2048)\}$.

block_out_channels $\in$ $\{(32, 128, 512), (128, 256, 512), (64, 256, 1024), (64, 512, 2048)\}$.

learning_rate $\in$ $\{1e\text{-}3, 5e\text{-}4, 1e\text{-}4, 1e\text{-}5\}$.

学习率 $\in$ $\{1e\text{-}3, 5e\text{-}4, 1e\text{-}4, 1e\text{-}5\}$。

train_batch_size $\in$ $\{8, 16, 64\}$.

训练批次大小 $\in$ $\{8, 16, 64\}$。

lr_warmup_steps $\in$ $\{100, 1000, 2000, 5000\}$.

lr_warmup_steps $\in$ $\{100, 1000, 2000, 5000\}$

D Experiments on Natural Data — Details

D 自然数据实验 - 细节

D.1 Model Configs

D.1 模型配置

```
layers_per_block = 2
block_out_channels = (512, 512, 1024)
```

down_block_types = (

down_block_types = (

"DownBlock2D",

下块2D

"CrossAttnDownBlock2D",

交叉注意力降维二维块

```
    "CrossAttnDownBlock2D",
)
up_block_types = (
    "CrossAttnUpBlock2D",
    "CrossAttnUpBlock2D",
    "UpBlock2D",
)
cross_attention_dim = 128
patch_size = 2
conv_in_kernel = 3
conv_out_kernel = 3
```

D.2 Training Configs

D.2 训练配置

Again there are two training stages. At the first stage the model learns to synthesize single-object crops. At the second stage the model learns to synthesize two objects in a specified spatial relation. After pretraining, we find it more efficient to only finetune weights in the projection and attention layers, without loss of performance compared to finetuning all parameters.

同样，训练分为两个阶段。在第一阶段，模型学习合成单个目标的裁剪。在第二阶段，模型学习在指定的空间关系下合成两个目标。预训练后，我们发现仅微调投影层和注意力层的权重更有效，与微调所有参数相比，性能没有损失。

Also, a larger finetuning learning rate leads to rapid convergence without hurting performance.

此外，较大的微调学习率会导致快速收敛，而不会损害性能。

Pretraining:

预训练:

```
image_size = (32,32)
```

```
mixed_precision = "fp16"
```

n o i s e _ s c h e d u l e = " squaredcos_cap_v2 "

noise_schedule = "squaredcos_cap_v2"

```
learning_rate = 1e-4
```

```
lr_warmup_steps = 1000
```

```
train_batch_size = 16
```

num_train_timesteps = 100 # d i f f u s i o n s t e p s

num_train_timesteps = 100 # diffusion steps

Finetuning:

微调：

```
image_size = (32,64) # H, W
```

```
mixed_precision = "fp16"
```

n o i s e _ s c h e d u l e = " squaredcos_cap_v2 "

noise_schedule = "squaredcos_cap_v2"

```
learning_rate = 5e-4
```

```
lr_warmup_steps = 1000
```

```
train_batch_size = 16
```

```
num_train_timesteps = 100 # diffusion steps
```

t r a i n a b l e _ p a r a m e t e r s = [" a t t e n t i o n s " , " encoder_hid_proj "]

可训练参数 = ["attentions", "encoder_hid_proj"]

D.3 Hyperparameter Search

D.3 超参数搜索

layers_per_block $\in$ {2, 3, 4}.

layers_per_block $\in$ {2, 3, 4}。

block_out_channels $\in$ {(64, 256, 1024), (128, 512, 1024), (512, 512, 1024), (512, 1024, 1536)}.

block_out_channels ∈ {(64, 256, 1024), (128, 512, 1024), (512, 512, 1024), (512, 1024, 1536)}。

learning_rate ∈ {1e-3, 5e-4, 1e-4}.

学习率 ∈ {1e-3, 5e-4, 1e-4}。

train_batch_size ∈ {16, 64}.

训练批次大小 ∈ {16, 64}。

lr_warmup_steps ∈ {1000, 2000, 5000, 10000}.

lr_warmup_steps ∈ {1000, 2000, 5000, 10000}

num_train_timesteps ∈ {100, 500, 1000}.

num_train_timesteps ∈ {100, 500, 1000}。

E Probing Experiments on Text Encodings

文本编码的探测实验

In the main paper, a conceptual framework for text-to-image generation is introduced, containing three components: text encoder, communication channel and image decoder. We have argued that the text-to-image generation performance crucially depends on all three components functioning and coordinating appropriately. Concretely, the text encoder is responsible for representing features extracted from raw text.

在主论文中，我们介绍了一个文本到图像生成的框架，该框架包含三个组件：文本编码器、通信通道和图像解码器。我们认为，文本到图像生成的性能关键取决于所有三个组件的正常运作和协调。具体来说，文本编码器负责表示从原始文本中提取的特征。

The communication channel aims at transmitting features cross-modality. Lastly, the image decoder instantiates an output into the pixel space. Note, though we conceptually divide up the pipeline into three components, **there is no hard boundary for the points at which different functions actually take place**, because all representations and transformations in this pipeline are homogeneously encoded as differentiable parameters.

该通信通道旨在跨模态传输特征。最后，图像解码器将输出实例化为像素空间。需要注意的是，虽然我们在概念上将管道划分为三个组件，但不同功能实际发生点的边界并不严格，因为此管道中的所有表示和转换都以可微参数的形式进行同质编码。

For example, a weak text encoder could under-process the text features, thus expecting the communication channel to further process text features on the fly. While it is convenient to consider three types of responsibilities for developing hypotheses regarding the sources of error, under the hood, these three components in an end-to-end system could easily transfer or take over responsibilities to or from each other.

例如，一个弱文本编码器可能会对文本特征进行欠处理，从而期望通信通道在运行时进一步处理文本特征。虽然考虑三种类型的责任来发展关于错误来源的假设很方便，但在底层，端到端系统中的这三个组件很容易互相转移或接管彼此的责任。

This raises the question: **How rich text features from pre-trained text encoders are?** As argued above, richer text features would give the subsequent communication channel an advantage point, lowering the complexity of transformations needed to transmit the key message cross-modality. This intuition aligns with the known benefit of representation learning, which saves efforts for downstream modeling.

这引发了一个问题：预训练文本编码器中的富文本特征有多丰富？如上所述，更丰富的文本特征将为后续的通信通道提供优势，降低跨模态传输关键信息的转换复杂度。这种直觉与表示学习的已知优势相一致，即为下游建模节省了工作量。

On the other hand, a less capable text encoder would increase the difficulty of learning the communication channel, and even create obstacles if critical information gets lost.

另一方面，能力较弱的文本编码器会增加学习通信通道的难度，甚至如果关键信息丢失，还会造成障碍。

Table A2: Probing results, showing that CLIP text encodings do not effectively facilitate subsequent modules to figure out the spatial positions of physical entities. In contrast, T5 and VLC are better candidates for coupling with text-to-image models.

表 A2：探测结果表明，CLIP 文本编码不能有效地帮助后续模块确定物理实体的空间位置。相比之下，T5 和 VLC 更适合与文本到图像模型耦合。

Configuration		Performance		
Text Encoder	MLP Num_layers	Tr Acc	Val Acc	Out-distr. Te Acc
CLIP-B/32	1	74.8	74.8	74.4
CLIP-B/32	2	74.6	74.4	74.2
CLIP-B/32	3	74.9	73.9	74.3
CLIP-L/14	1	74.6	73.5	74.3
CLIP-L/14	2	74.8	74.3	74.6
CLIP-L/14	3	75	74.9	74.5
T5-small	1	98.5	98.5	93.1
T5-small	2	100	100	95.8
T5-small	3	100	100	96.7
T5-flan-xxl	1	99.9	99.8	99.3
T5-flan-xxl	2	100	100	99.8
T5-flan-xxl	3	100	100	99.4
VLC-L/16	1	99.1	99.1	96.1
VLC-L/16	2	99.9	100	96.0
VLC-L/16	3	100	100	99.8

To this end, we perform probing experiments to analyze how good an advantage point a text encoder can provide for learning a communication channel. We consider three pre-trained text encoders: CLIP [33], T5 [34] and VLC [9], which are chosen to be representative of different model families and pre-training objectives.

为此，我们进行了探测实验，以分析文本编码器在学习通信信道方面能提供多好的优势点。我们考虑了三种预训练的文本编码器：CLIP [33],、T5 [34] 和 VLC [9],，它们被选为不同模型族和预训练目标的代表。

CLIP is an encoder-only dual architecture pre-trained with image-text contrastive loss. T5 is an encoder-decoder architecture pre-trained with the causal language modeling loss. VLC is an encoder-only multimodal architecture pre-trained with masked language modeling, masked image modeling as well as image-text matching objectives.

CLIP 是一种仅编码器的双重架构，使用图像-文本对比损失进行预训练。T5 是一种编码器-解码器架构，使用因果语言建模损失进行预训练。VLC 是一种仅编码器的多模态架构，使用掩码语言建模、掩码图像建模以及图像-文本匹配目标进行预训练。

Pertaining to our work, the core responsibility of the communication channel is to translate linguistic fillers and roles to the corresponding visual entities and spatial positions. Thus, we train probing classifiers to label the noun tokens in a caption with spatial positions (e.g. top or bottom).

就我们的工作而言，通信通道的核心职责是将语言填充物和角色转换为相应的视觉实体和空间位置。因此，我们训练探测分类器，以使用空间位置（例如，顶部或底部）标记标题中的名词标记。

For example, given a caption “a book is on top of a cup”, we apply an MLP on the encoded features of the “book” and “cup” tokens, and expect the MLP to output label 0 for “book” and label 1 for “cup”.

例如，给定一个标题“a book is on top of a cup”，我们对“book”和“cup”词元的编码特征应用一个多层感知机（MLP），并期望MLP对“book”输出标签0，对“cup”输出标签1。

For another example, “a laptop is at the bottom of a phone”, the MLP should output label 1 for “laptop” and label 0 for “phone”. The probing performance signifies the utility of a pre-trained text encoder in assisting a diffusion model, particularly in the context of generating spatial relations.

例如，对于“笔记本电脑在手机底部”这句话，**MLP** 应该为“笔记本电脑”输出标签 1，为“手机”输出标签 0。探测性能表明预训练文本编码器在辅助扩散模型，特别是在生成空间关系.方面具有实用价值。

Though a probing classifier is a drastic simplification, we believe that it still captures the main problem.

尽管探测分类器是一个极大的简化，但我们相信它仍然捕捉到了主要问题。

We select 200 common English nouns meaning concrete objects, then divide them into 170 “in-distribution” and 30 “out-of-distribution” nouns. We expect a successful probe to generalize to out-of-distribution nouns, which would strongly imply the existence of robust feature directions encoding spatial information.

我们选取了 200 个常见的英语名词，这些名词指的是具体物体，然后将它们分为 170 个“分布内”名词和 30 个“分布外”名词。我们期望一个成功的探测器能够泛化到分布外名词，这将强烈暗示存在着编码空间信息的鲁棒特征方向。

We train 1-, 2-, and 3-layer MLPs with ReLU nonlinearities. The 1-layer MLP is equivalent to linear probing. The MLPs are trained for 10-20 epochs until convergence. We observe consistent results across random data-splitting outcomes (5 seeds) and hyperparameters such as learning rate ({1e-3, 1e-4}) and batch size ({16, 64}).

我们训练了具有 ReLU 非线性的 1 层、2 层和 3 层 MLP。1 层 MLP 等效于线性探测。MLPs 训练 10-20 个 epoch 直到收敛。我们观察到在随机数据分割结果（5 个种子）和超参数（如学习率（{1e-3, 1e-4}）和批次大小（{16, 64}））之间的一致结果。

Table A2 summarizes our results. CLIP struggles to exceed 75% and this cannot be improved by increasing the encoder size or the number of MLP layers. T5-small achieves perfect in-distribution performance, but makes minor mistakes when generalizing out-of-distribution. T5-flan-xxl [5] achieves perfect accuracy in all three splits.

表 A2 总结了我们的结果。CLIP 难以超过 75%，并且通过增加编码器大小或 MLP 层数无法改善。T5-small 在分布内实现了完美的性能，但在分布外泛化时会犯一些小错误。T5-flan-xxl [5] 在所有三个拆分中都实现了完美的准确率。

VLC performs on par with T5-flan-xxl indistribution, and slightly underperforms T5-flan-xxl out-of-distribution.

VLC 在分布内与 T5-flan-xxl 表现相当，而在分布外略逊于 T5-flan-xxl。

Our probing experiments undermine CLIP-text-encoder's effectiveness in facilitating subsequent modules to figure out the spatial positions of physical entities. In contrast, T5 and VLC are better candidates for coupling with text-to-image models. Indeed, our ablation study on diffusion models confirms this claim.

我们的探测实验表明，CLIP 文本编码器在帮助后续模块确定物理实体的空间位置方面效果不佳。相比之下，T5 和 VLC 更适合与文本到图像模型结合。事实上，我们对扩散模型的消融研究证实了这一说法。

As shown in Figure A5, all experiments follow the same setting except that solid lines represent the T5-small text encoder and dotted lines represent the CLIP-B/16 text encoder. It is evident that dotted lines consistently fall behind solid lines. Note, **our analysis does not render CLIP as incapable of serving as the text encoder for text-to-image models.**

如图 A5 所示，所有实验遵循相同的设置，不同之处在于实线代表 T5-small 文本编码器，虚线代表 CLIP-B/16 文本编码器。很明显，虚线始终落后于实线。需要注意的是，我们的分析并没有得出 CLIP 不适合作为文本到图像模型的文本编码器的结论。

Nevertheless, compared to T5 and VLC, CLIP might have to offload more burden to the communication channel for nontrivial processing.

然而，与**T5**和**VLC**相比，**CLIP**可能需要将更多非平凡处理的负担转移到通信通道。

F Ablation Results on Image Positional Embeddings

图像位置嵌入的消融结果

In this section, we investigate a crucial factor — image positional embeddings — that draws a connection from spatial features to the correct absolute positions in the 2D image coordinates. Without explicitly embedding patch positions, such ability could be largely damaged.

在本节中，我们研究了一个关键因素——图像位置嵌入——它将空间特征与二维图像坐标中的正确绝对位置联系起来。如果没有显式地嵌入补丁位置，这种能力可能会受到很大损害。

Though, arguably, zero-paddings in the convolutional layers could provide complementary positional information, we believe that this cannot fully compensate for the loss incurred by a lack of explicit image positions. **We perform ablation studies where the image positional embeddings are removed while remaining other**

architectural components intact. Specifically, we set the number of concepts to be 30 because a smaller number of concepts imposes a greater challenge such that discrepancies in performance can be easily revealed.

尽管卷积层中的零填充可以提供补充的位置信息，但我们认为这并不能完全弥补缺乏显式图像位置带来的损失。我们进行了消融研究，其中图像位置嵌入被移除，而其他架构组件保持完整。具体来说，我们将概念数量设置为 30，因为较少的概念数量会带来更大的挑战，因此性能差异可以很容易地显现出来。

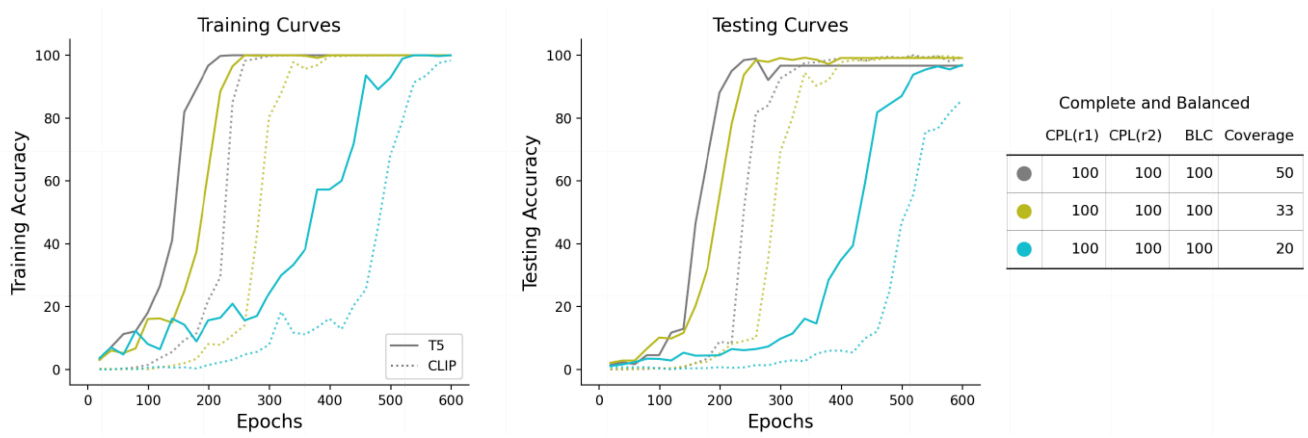


Fig. A5: Ablation study on the text encoder while keeping all other training configurations fixed. The performance of diffusion paired with CLIP text encoder consistently falls behind the performance with T5, suggesting that CLIP would indeed make learning harder and slower.

图 A5：在保持所有其他训练配置不变的情况下，对文本编码器进行消融研究。扩散模型与 CLIP 文本编码器配对的性能始终落后于 T5 的性能，这表明 CLIP 确实会使学习变得更加困难和缓慢。

Figure A6 presents the results of this ablation study. For notational convenience, we call models *with* image positional embeddings the “Pos-models", and call models *without* image positional embeddings the “NoPos-models". It can be observed that, NoPosmodels perform worse than their counterpart Pos-models across all panels.

图A6展示了该消融研究的结果。为了方便表示，我们将具有图像位置嵌入的模型称为“Pos-模型”，将没有图像位置嵌入的模型称为“NoPos-模型”。可以观察到，NoPos-模型在所有面板上的表现都比其对应的Pos-模型差。

Comparing the panels between rows, we identify different types of failure. In the first row (pink and grey), NoPos-models exhibit much slower convergence speed, but are still able to converge and generalize eventually. In the second row (purple and brown), NoPosmodels do not fall behind in terms of training accuracy.

比较各行面板，我们识别出不同类型的故障。在第一行（粉红色和灰色）中，NoPos 模型表现出更慢的收敛速度，但最终仍能收敛并泛化。在第二行（紫色和棕色）中，NoPos 模型在训练精度方面没有落后。

Yet they are generalizing to the testing set much worse than Pos-models. In the third row (green and red), there are small gaps between Pos-models and NoPos-models in training accuracy. And this gap widens regarding testing accuracy. In the fourth row (blue and orange), unfortunately, NoPos-models have trouble even fitting the training set.

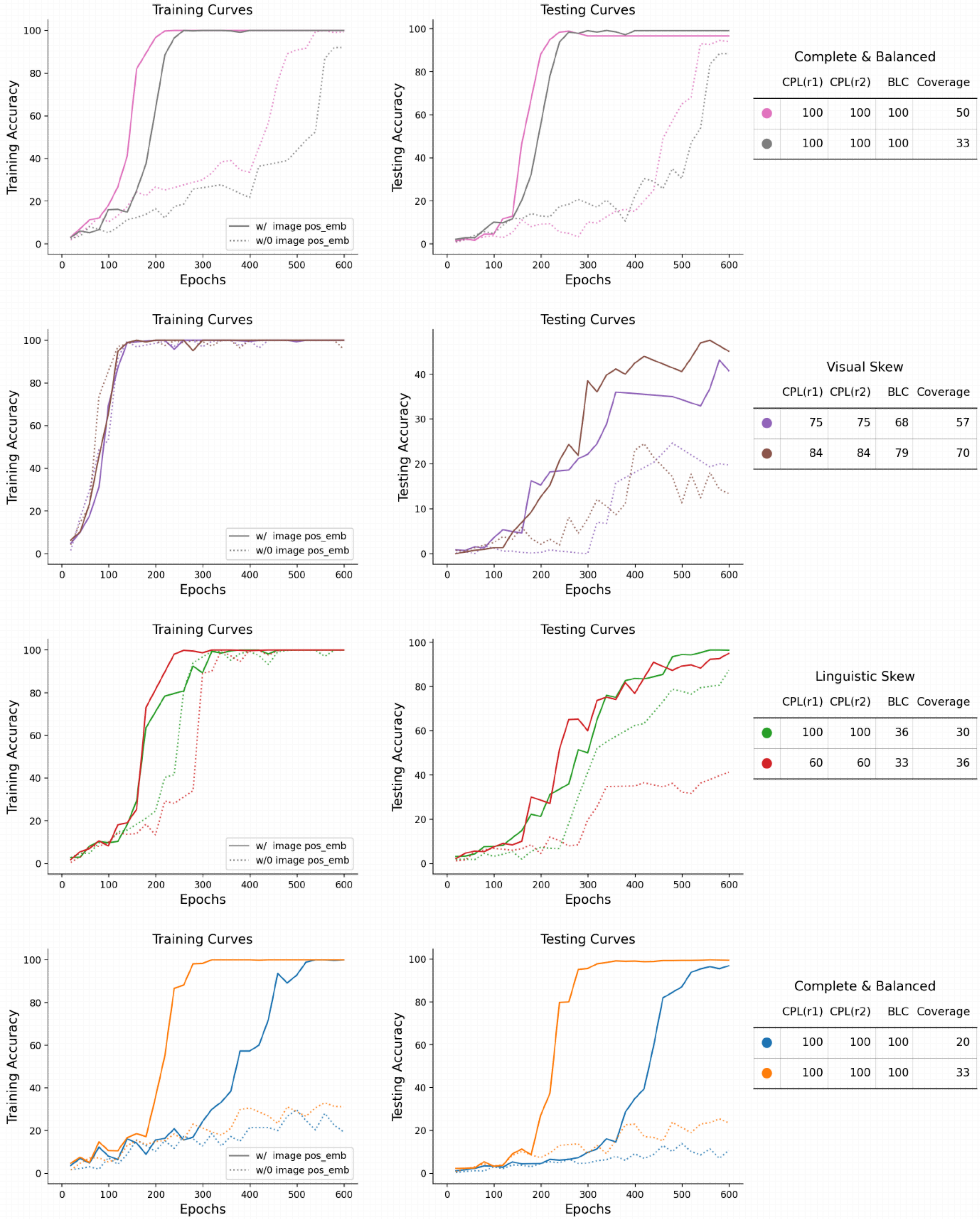
然而，它们在测试集上的泛化能力远不如Pos模型。在第三行（绿色和红色）中，Pos模型和NoPos模型在训练精度上存在微小差距。而这个差距在测试精度方面会扩大。在第四行（蓝色和橙色）中，不幸的是，NoPos模型甚至难以拟合训练集。

We attempt to associate the failure patterns of NoPos-models with our proposed linguistic or visual skew in the training data. It appears that in rows 2&3, models are hampered by the skew, with NoPos-models being more severely hampered. When the training data is complete and balanced, the NoPos-models would generalize on larger coverages (row 1), albeit at much slower paces, and fail spectacularly on smaller coverages (row 4).

我们试图将 NoPos 模型的失败模式与我们提出的训练数据中的语言或视觉偏差联系起来。结果表明，在第 2 行和第 3 行中，模型受到了偏差的影响，其中 NoPos 模型受到的影响更为严重。当训练数据完整且平衡时，NoPos 模型将在更大的覆盖范围（第 1 行）上进行泛化，尽管速度要慢得多，并且在较小的覆盖范围（第 4 行）上会发生严重失败。

We suspect that the interplay between the data skew and the failure pattern of NoPos-models is intricate, so we have not reached a conclusive statement. But the key takeaway is clear: **removing image positional embeddings leads to notable performance degradation in all cases**.

我们怀疑数据倾斜和 NoPos 模型的故障模式之间的相互作用错综复杂，因此我们还没有得出结论性的结论。但关键的结论很明确：在所有情况下，移除图像位置嵌入都会导致显著的性能下降。



Complete & Balanced

	CPL(r1)	CPL(r2)	BLC	Coverage
	100	100	100	50
	100	100	100	33

Visual Skew

	CPL(r1)	CPL(r2)	BLC	Coverage
	75	75	68	57
	84	84	79	70

Linguistic Skew

	CPL(r1)	CPL(r2)	BLC	Coverage
	100	100	36	30
	60	60	33	36

Complete & Balanced

	CPL(r1)	CPL(r2)	BLC	Coverage
	100	100	100	20
	100	100	100	33

Fig. A6: Ablation study on whether to include image positional embeddings. We see that diffusion models without the image positional embedding are inferior to their counterparts across all panels, albeit failing or underperforming for different reasons.

图 A6：关于是否包含图像位置嵌入的消融研究。我们看到，没有图像位置嵌入的扩散模型在所有面板中都比它们的对应模型差，尽管它们由于不同的原因而失败或表现不佳。

G Latent Diffusion Experiments

G 潜扩散实验

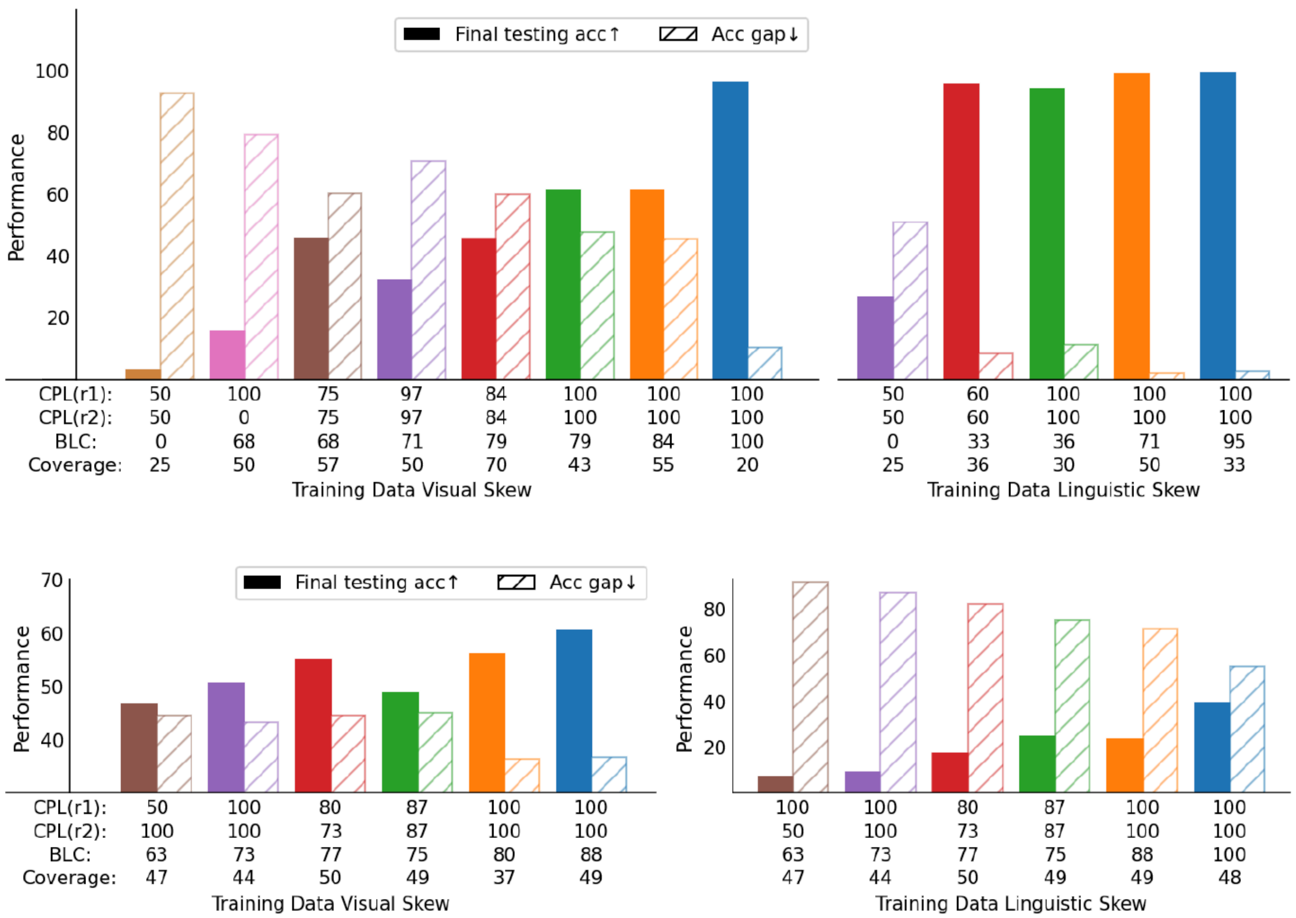


Fig. A7: Results for latent diffusion models. A pretrained VAE is frozen, while the UNet is trained from scratch. On both synthetic (top) and natural (bottom) images, we observe trends consistent with our pixel diffusion findings — lower CPL and BLC harms testing accuracy and widens the generalization gap, while Coverage is less predictive.

图 A7: 潜在扩散模型的结果。预训练的 VAE 被冻结，而 UNet 从头开始训练。在合成图像（顶部）和自然图像（底部）上，我们观察到与像素扩散发现一致的趋势——较低的 CPL 和 BLC 损害测试准确性并扩大泛化差距，而覆盖率的预测性较低。

We repeat the experiments on synthetic and natural images with a latent diffusion model, in order to validate that our arguments do not break when moving from the pixel to a latent space. We adopt a pretrained VAE (stabilityai/stable-diffusion-2-1), freeze it, and train the UNet from scratch. We use the same set of hyperparameters as in Appendix C and D, except for two modifications: 1) input resolution is increased by a factor of four, 2) we use four channels to match the input/output dimensions of the VAE, as opposed to three for rgb inputs.

我们使用潜在扩散模型在合成和真实图像上重复了实验，以验证我们的论点在从像素空间迁移到潜在空间时不会失效。我们采用了一个预训练的 VAE (stabilityai/stable-diffusion-2-1)，将其冻结，并从头开始训练 UNet。我们使用与附录 C 和 D 中相同的超参数集，除了以下两个修改：1) 输入分辨率增加了四倍，2) 我们使用四个通道来匹配 VAE 的输入/输出维度，而不是 RGB 输入的三个通道。

Results are plot in Figure A7, which are highly consistent with our pixel diffusion findings.

结果绘制在图 A7 中，与我们的像素扩散发现高度一致。

H Auto-Evaluation of Relations

关系自动评估

Our experiments critically depend on an auto-eval engine that correctly judges both the object identity and their spatial relations. However, current evaluation methods of generated images have lingering issues of not paying attention to relations. Off-the-shelf open-vocabulary detection models have problems such as insensitivity to certain object classes, redundant predictions and poor reliability at low resolutions.

我们的实验严重依赖于一个自动评估引擎，该引擎能够正确判断物体身份及其空间关系。然而，当前生成的图像评估方法存在着忽视关系的长期问题。现成的开放词汇检测模型存在诸如对某些物体类别不敏感、预测冗余以及在低分辨率下可靠性差等问题。

Thus, we decide that they are not suitable for our purpose. We then try to leverage vision-language models, inspired by [13, 26], which ended up not working. Eventually, we arrive at a working solution via finetuning ViT [6], despite making certain compromises, i.e. assuming that objects only occur at fixed image regions. **This section describes our attempts at prompting VLMs and finetuning ViT for automatic evaluation, pertaining to experiments on natural images.**

因此，我们认为它们不适合我们的目的。然后，我们尝试利用视觉语言模型，受 [13, 26] 的启发，但最终没有成功。最终，我们通过微调 ViT [6]，得到了一个可行的解决方案，尽管做出了一些妥协，即假设物体只出现在固定的图像区域。本节描述了我们尝试使用 VLM 和微调 ViT 进行自动评估的尝试，涉及对自然图像. 的实验。

H.1 Leveraging Large Vision-Language Models

H.1 利用大型视觉语言模型

Zero-Shot Multiple-Choice VQA Our first attempt was using off-the-shelf VLMs — BLIP2 [20] and LLaVA1.5 [24] — to evaluate spatial relations, since they have been trained on position-related data (e.g. Visual Genome [18]). We phrase the prompt as a zero-shot multiple-choice question. For example,

零样本多选VQA 我们的首次尝试是使用现成的 VLM——BLIP2 [20] 和 LLaVA1.5 [24]——来评估空间关系，因为它们已经在与位置相关的数据（例如 Visual Genome [18]）上进行了训练。我们将提示语表述为一个零样本多选问题。例如，

Which c a p t i o n b e t t e r d e s c r i b e s the image ?

哪一个标题更能描述这张图片?

A. A mug to the r i g h t o f a k n i f e ;

A. 一把在刀子右边的马克杯;

B. A mug i n f r o n t o f a k n i f e ;

B. 一把刀前的马克杯;

C. A mug behind a k n i f e ;

C. 刀后边的马克杯;

D. A mug to the l e f t o f a k n i f e .

D. 一把在刀子左边的马克杯。

Table A3 shows that both LLaVA1.5-VQA and BLIP2-VQA barely outperformed 25% on the 4-way classification task.

表 A3 显示，LLaVA1.5-VQA 和 BLIP2-VQA 在 4 类分类任务中，其表现都略微超过 25%。

Table A3: Zero-Shot Multiple-Choice VQA Results of BLIP2 and LLaVA1.5 on the What's Up Benchmark.

表 A3：BLIP2 和 LLaVA1.5 在 What's Up 基准测试上的零样本多项选择 VQA 结果。

Model	Accuracy
BLIP2	0.26
LLaVA1.5	**0.28**

Table A4: Zero-Shot Object Detection Results of BLIP2 and LLaVA1.5 on the What's Up Benchmark.

表 A4：BLIP2 和 LLaVA1.5 在 What's Up 基准测试上的零样本目标检测结果。

	BLIP2	LLaVA1.5
Acc. on True Labels	**100%**	83.6%
Acc. on False Labels	12.1%	**94.2%**
Average Accuracy	16.9%	**93.6%**

Zero-shot Object Detection Though VLMs struggle to recognize positional relationships, they do excel in detecting the existence of objects. VLMs could still be helpful if the generated images were preprocessed into single-object crops, under the assumption that objects only occur in fixed image regions and cropping does not cut through objects.

尽管 VLM 在识别位置关系方面存在困难，但它们在检测物体存在方面表现出色。如果生成的图像被预处理成单物体裁剪，在物体只出现在固定图像区域且裁剪不会穿过物体的假设下，VLM 仍然会有所帮助。

For example, in our setting with binary spatial relations and a 32x64 input size, preprocessing involves cropping the image into two 32x32 squares. We then feed the two crops into the VLM in two individual passes, together with the prompt: "`Is a [object] in the image?`", where we loop through all 18 object classes in the What'sUp benchmark.

例如，在我们的二元空间关系设置中，输入大小为 32x64，预处理涉及将图像裁剪成两个 32x32 的正方形。然后，我们将这两个裁剪后的图像分别输入 VLM，并附带提示："图像中是否有 [对象]？"，其中我们循环遍历 What'sUp 基准中的所有 18 个对象类别。

As shown in Table A4, BLIP2 tends to affirm the presence of any queried object, leading to 100% accuracy for true labels but a significantly worse accuracy, 12.1%, for false labels, indicating a high false-positive rate. Given that there are 18 objects in total, questions with false labels outnumber those with true labels by 17 times.

如表 A4 所示，BLIP2 倾向于肯定任何查询对象的出现，导致真标签的准确率为 100%，但假标签的准确率显著下降至 12.1%，表明存在较高的假阳性率。鉴于共有 18 个对象，带有假标签的问题比带有真标签的问题多 17 倍。

This explains why the average accuracy is close to that for false labels.

这解释了为什么平均准确率接近于错误标签的准确率。

In contrast, LLaVA1.5 remarkably outperforms BLIP2 with much more balanced false-positive and false-negative rates. Therefore, we proceed to compute the frequency when LLaVA judges correctly for both crops given a generated image. We find certain object classes particularly difficult for LLaVA.

相比之下，LLaVA1.5 在平衡误报率和漏报率方面显著优于 BLIP2。因此，我们继续计算在给定生成图像的情况下，LLaVA 对两个裁剪图像都正确判断的频率。我们发现某些物体类别对 LLaVA 来说特别困难。

The overall accuracy is 68.14%. Excluding difficult classes progressively improves accuracy to 92% (Table A5). However, we are not satisfied with LLaVA as an auto-eval engine, because we expect near-oracle accuracy. We list the top10 frequent misclassifications in Table A6.

整体准确率为 68.14%。排除困难类别逐步提高准确率至 92%（表 A5）。然而，我们对 LLaVA 作为自动评估引擎并不满意，因为我们期望接近先知的准确率。我们在表 A6 中列出了前 10 个最常见的错误分类。

Understandably, LLaVA tends to confuse between semantically similar objects.

可以理解的是，LLaVA 容易混淆语义相似的物体。

Table A5: LLaVA1.5 Zero-Shot Object Detection has highly unbalanced accuracy between easy and hard classes

表 A5：LLaVA1.5 零样本目标检测在简单类别和困难类别之间存在高度不平衡的准确率。

Inclusion of classes	Accuracy
All	0.68
Exclude the worst 2 Classes (Tape, Flower)	0.83
Exclude the worst 3 Classes (Tape, Flower, Knife)	0.87
Exclude the worst 4 Classes (Tape, Flower, Knife, Fork/Scissors)	0.89
Exclude the worst 5 Classes (Tape, Flower, Knife, Fork, Scissors)	0.92

Table A6: LLaVA1.5 Zero-Shot Object Detection: Top 10 frequent misclassifications

表 A6：LLaVA1.5 零样本目标检测：前 10 个最常出现的误分类

(Correct, Predicted)	Frequency
(Mug, Cup)	68
(Can, Cap)	36
(Bowl, Plate)	28
(Sunglasses, Headphones)	27
(Remote, Headphones)	27
(Scissors, Remote)	25
(Scissors, Headphones)	24
(Scissors, Cap)	21
(Flower, Candle)	21
(Candle, Cap)	19

H.2 Crop and Classify

H.2 裁剪和分类

For the purpose of automatically evaluating the generated images on the What'sUp benchmark, we finally arrive at a working solution that consists of two steps. The first step obtains single object crops from an image based on heuristics such that the dividing line between two objects aligns with the center of the image.

为了自动评估What'sUp基准测试中生成的图像，我们最终得出一个包含两个步骤的工作解决方案。第一步根据启发式方法从图像中获取单个目标裁剪，使得两个目标之间的分界线与图像中心对齐。

The second step classifies the object into one of the 15 classes. We obtain an Oracle classifier via finetuning ViTB/16[11]. We use individual object crops from the original images as our finetuning data, splitting into training and validation sets with the ratio 90:10.

第二步将物体分类到15个类别中的一个。我们通过微调ViTB/1611获得了一个Oracle分类器。我们使用原始图像中的单个物体裁剪作为我们的微调数据，并以90:10的比例将其分成训练集和验证集。

Examples are illustrated in Figure A8. We adopt the Adam optimizer with lr=2e-4 and a batch size of 16. The classification accuracy reaches perfect within 5 epochs. For sanity check, we manually examined 144 generated examples and our ViT evaluated correctly on all of them.

示例如图 A8 所示。我们采用 Adam 优化器，学习率 lr=2e-4，批次大小为 16。分类精度在 5 个 epoch 内达到完美。为了进行健全性检查，我们手动检查了 144 个生成的示例，我们的 ViT 对所有示例都进行了正确评估。

Fig. A8: Finetuning data examples. We finetune ViT to classify objects in image crops for the purpose of automatically evaluating generated images.

图 A8: 微调数据示例。我们微调 ViT 以对图像裁剪中的物体进行分类，目的是自动评估生成的图像。

I Limitations and Future Work

I. 限制与未来工作

This work bridges a significant gap by formally characterizing the phenomenological space in text-to-image generation datasets. Though we mainly focus on small-scale and controllable settings for proof-of-concept experiments, a breadth of expansions are enabled as direct next steps.

这项工作通过正式刻画文本到图像生成数据集中的现象学空间，弥合了重要的差距。虽然我们主要关注小规模和可控环境以进行概念验证实验，但作为直接的下一步，可以进行广泛的扩展。

Larger-scale data The scaling property of our argument remains to be testified in future work. Current difficulties in scaling up lie in the lack of clean data with entity-relation annotations. Controlled experiments require datasets spanning the incomplete $\rightarrow$ complete and imbalanced $\rightarrow$ balanced spectra.

更大规模的数据 我们论点的可扩展性有待于在未来的工作中得到验证。目前扩展的困难在于缺乏带有实体-关系标注的干净数据。受控实验需要跨越不完整 $\rightarrow$ 完整和不平衡 $\rightarrow$ 平衡频谱的数据集。

One path is to draw subsamples, filtering with relations of interest and for objects bound to different roles. We found Flickr and Nocaps too small after filtering. As seen in Table A1, the number of images drops to 16k and 2k for Flickr and Nocaps, respectively, for the left-right relation.

一种方法是抽取子样本，使用感兴趣的关系进行过滤，并针对绑定到不同角色的对象进行过滤。我们发现 Flickr 和 Nocaps 在过滤后太小了。如表 A1 所示，对于左右关系，Flickr 和 Nocaps 的图像数量分别下降到 16k 和 2k。

These will further drop if we desire a high balance score. Our work motivates the development of either cleaner datasets or augmentation techniques to study with perceptually rich images.

如果我们希望获得更高的平衡分数，这些分数将进一步下降。我们的工作促使开发更干净的数据集或增强技术，以便使用感知丰富的图像进行研究。

Complex relations There is ample room for future study to explore beyond binary, grammatical or spatial relations. We see three possible directions: 1) study relations involving >2 roles, e.g. “forming a line/triangle"; 2) distinguish the directionality of a relation, and extend to non-directional relations, e.g. “next to"; 3) expand to interaction-based relations, usually entailed by verbs, e.g. “serve", “feed", “approach" — more abstract and dynamic relations may require other computer vision advances such as action/posture recognition, as well as incorporating a temporal axis.

复杂关系 未来研究有很大的空间去探索超出二元、语法或空间关系的范围。我们看到三个可能的方向：1）研究涉及>2个角色的关系，例如“形成一条线/三角形”；2）区分关系的方向性，并扩展到非方向性关系，例如“旁边”；3）扩展到基于交互的关系，通常由动词蕴含，例如“服务”、“喂食”、“接近”——更抽象和动态的关系可能需要其他计算机视觉方面的进步，例如动作/姿势识别，以及结合时间轴。

Guiding data pruning Establishing links between data distributional properties and models' generalizing behavior not only benefits model understanding and development, but also potentially inspires data pruning techniques to save compute and obtain better performance. In fact, experiments in Section 5 exhibit such a flavor: we did not use the full What'sUp dataset containing 18 objects, but discarded frequent objects that worsen the skew.

引导数据剪枝 建立数据分布特性与模型泛化行为之间的联系，不仅有利于模型理解和开发，而且有可能激发数据剪枝技术，以节省计算量并获得更好的性能。事实上，

第 5 节中的实验展示了这种趋势：我们没有使用包含 18 个对象的完整 What'sUp 数据集，而是丢弃了导致数据倾斜的频繁对象。

Future work may develop pruning strategies informed by phenomenological measurements and evaluate the data-efficiency gain.

未来的工作可以开发基于现象学测量的修剪策略，并评估数据效率的提升。